Introdução ao conceito de número

Dialéticas entre finito/infinito, discreto/contínuo e relativo/absoluto

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Projeto (A)simetrias

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General References

- L. Corry, *A Brief History of Numbers*, 2015, Oxford University Press.
- F. Patras, *The Essence of Numbers*, 2020, Springer.
- A. Badiou, *Le Nombre et les nombres*, 1990, Seuil.

What is mathematics?

Philosophical Positions about Mathematics

- « **Platonisms** » (Plato, Gödel, Badiou, Lautman, Connes): mathematics is about entities that exist independently of any cognitive act.

- **Aristotelism**: mathematics is about structures and patterns abstracted or idealized from experience.

- **Logicism** (Frege, Russell): mathematics can be reduced to logic, which encodes the « laws of thought ».

- **Formalism** (Peano, Hilbert): mathematics is a « language game » given by the manipulation of signs according to certain rules.

- **Intuitionism** (Brouwer): mathematics results from the constructive rational activity of the human mind.

- **Structuralism**: mathematics is the study of objects defined as « positions in a relational structure ».

1. Mathematical « Platonism »

Mathematics is a science that studies a particular realm of entities among others — that of abstract, *ideal* or *supersensible objects* —, which exist in themselves in the sense that they are independent of the cognitive activity of the mathematicians (e.g. Gödel).

« [...] the objects and theorems of mathematics are as objective and independent of our free choice and our creative acts as is the physical world. [...] the Platonistic view is the only one tenable. Thereby I mean the view that mathematics describes a non-sensual reality, which exists independently both of the acts and [of] the dispositions of the human mind and is only perceived, and probably perceived very incompletely, by the human mind. »

K. Gödel, *Some basic theorems on the foundations of mathematics and their implications*.

« Platonists will be those who consider mathematics to be the *discovery* of truths about structures that exist independently of the activity or thought of mathematicians. »

P. Benaceraff and H. Putnam, Introduction to *Philosophy of Mathematics*, p. 18.

A. Connes sur la « réalité mathématique archaique »

« A. Connes: J'avoue être résolument platonicien […] je maintiens qu'elle sont un objet, tout aussi réel que celui des sciences dont je parlais plus haut, mais qui n'est pas matériel, et n'est localisé ni dans l'espace, ni dans le temps. Il a cependant une existence tout aussi ferme que la réalité extérieure et les mathématiciens s'y heurtent un peu comme on se heurte à un objet matériel dans la réalité extérieure. Cette réalité dont je parle, du fait qu'elle n'est localisable ni dans l'espace ni dans le temps, donne, lorsqu'on a la chance d'en dévoiler une infime partie, une sensation de jouissance extraordinaire par le sentiment d'intemporalité qui s'en dégage.

M.P. Schützenberger: […] thèse léniniste: la matière c'est ce qui résiste, les objets mathématiques résistent, donc les objets mathématiques ont au moins autant d'existence que la matière… »

A. Connes, A. Lichnerowicz et M.P. Schützenberger, *Triangle de pensées*, Paris: Odile Jacob, 2000, pp. 37 & 39.

Against the « vulgar » understanding of Platonism

« In the open debate between formalist and intuitionist, since the discovery of the transfinite, mathematicians have become accustomed to summarily designate under the name Platonism any philosophy for which the existence of a mathematical entity is taken as assured, even though this entity could not be built in a finite number of steps. It goes without saying that this is a superficial knowledge of Platonism, and that we do not believe ourselves to be referring to it. All modern Plato commentators on the contrary insists on the fact that Ideas are not immobile and irreducible essences of an intelligible world, but that they are related to each other according to the schemas of a superior dialectic that presides over their arrival. »

A. Lautman, *Essay on the Notions of Structure and Existence in Mathematics.*

Objection to mathematical « Platonism »

This formulation of « Platonism » does not explain

« The Unreasonable Effectiveness of Mathematics

in the Natural Sciences » (E. Wigner).

That is, it does not explain its transversality

with respect to different domains of entities

(e.g. physics, biology, economics).

The Threefold Doctrine of Science

1. Sciences *qua* theories of « life »

The different sciences

Physics,
Chemistry,
Biology,
Geology
Economy,
Sociology,
etc.

study the different dimensions or « ontic regions » (Heidegger)

of « life » or « nature ».

The Threefold Doctrine of Science

II. Mathematics *qua* Theory of Science

Mathematics is not so much a science among others

as a scientific discipline that studies

the formal structures of rational thinking as such,

that is, a ***science of science***.

For Kant,

mathematics is not so much a science

as the quintessence or the signature of scientificity as such.

« […] in any special doctrine of nature there can be only as much *proper* science as there is *mathematics* therein. »

(E. Kant, *Metaphysical Foundations of Natural Science*)

« A famous query of Eugene Wigner asks for an explanation of the unreasonable effectiveness of mathematics—especially its effectiveness in the various sciences. […] One might say that mathematics, too, is a science; namely it is that science whose objects are the forms or structures which are common to different specific scientific realizations. I am inclined to restate this by asserting that mathematics is protean science; its subject matter consists of those structures which appear (unchanged) in different scientific contexts. »

S. Mac Lane, *Structure in Mathematics*, 1996.

« *Τα μαθηματα* means, in Greek, that which, in his observation of beings and interaction with things, man knows in advance. »

M. Heidegger, *The Age of the World Picture*, in *Off The Beaten Track*.

This characterization of mathematics

as a ***science of science***

can be rephrased :

- in ***ontological*** terms (cf. Badiou).

- in ***transcendental/speculative*** terms (cf. Frege).

Mathematics *qua* Ontology (Badiou)

« Mathematics *is* ontology, the science of being qua being. »

A. Badiou, *Being and Event*, p.4.

« […] in terms of the 'application' of mathematics to the so-called natural sciences […] the clarification is immediately evident if mathematics is the science, in any case, of everything that is, *insofar as it is*. »

A. Badiou, *Being and Event*, p.7.

In the wake of Badiou,

mathematics can be understood as the science

that studies the most general categories

of rational thought,

that is, those that can be applied to anything

simply by virtue of the fact that it is.

Ontological categories

correlated to different branches of mathematics

- **Multiplicity** (*being-many*): set theory.
- **Identity** (*being-itself*): group theory.
- **Relation** (*being-with*): set-theoretic relations, category theory.
- **Extension** (*being-there*): geometry.
- **Quality** (*being-thus*): category theory.
- **Quantity** (*being-thus-much*): arithmetics, number theory, differential and integral calculus.
- **Difference** (*being-different*): category theory, homotopy type theory.
- **Continuity** (*being-unbreakable*): differential and integral calculus.

Mathematics *qua* speculative construction of rational tools (Frege)

Mathematics is an activity by means of which

we can refine and enhance

the rational faculties and categories

of human understanding in order to perform specialized tasks.

According to Frege's analogies,

mathematical logic is to ***human language***

what a ***microscope*** is to the ***eye***

or what a ***robotic hand*** is to the ***human hand***.

« I believe that I can best make the relation between my ideography to ordinary language clear if I compare it to that which the microscope has to the eye. Because the range of its possible uses and the versatility with which it can adapt to the most diverse circumstances, the eye is far superior to the microscope. Considered as an optical instrument, to be sure, it exhibits many imperfections, which ordinarily remain unnoticed only on account of its intimate connection with our mental life. But, as soon as scientific goals demand great sharpness of resolution, the eye proves to be insufficient. The microscope, on the other hand, is perfectly suited to precisely such goals, but that is just why it is useless for all others. »

G. Frege, *Begriffsschrift*, Preface.

« Language [...] proves to be deficient in the matter of protecting our thinking from error; it does not even fulfill the first demand made on it in this respect, that of being unequivocal. [...] Language is not in such a way dominated by logical laws that compliance with grammar would of itself guarantee the correctness of thought processes. [...] The shortcomings here stressed are caused by a certain softness and instability of language which, on the other hand, constitute the reason for its many-sided usefulness and potentiality for development. In this respect language can be compared to the hand which, despite its adaptability to the most diverse tasks, is still inadequate. We build ourselves artificial hands-tools for special purposes which function more exactly than the hand is capable of doing. And how is this exactness possible? Through the very rigidity and inflexibility of the parts, the lack of which makes the hand so dexterous. The word language is in the same way also not adequate. We have need of a system of symbols from which every ambiguity has been banned and whose rigorous logical form cannot be escaped by the content. »

G. Frege, *On the Scientific Justification of a Concept-Script*.

To do science means

to try to understand a given « ontic domain »

- by thinking logically (logic);

- by methodically « forgetting » features that are contextually irrelevant (abstraction);

- by quantifying by means of numbers (order structures, algebraic structures, etc.);

- by combining and computing (algebra);

- by predicting (differential and integral calculus);

- by using spatial schematizations (geometry):

- by identifying patterns, symmetries and invariants (group theory).

We can understand the different branches of mathematics

as conceptual and formal technologies

by means of which we can enhance

the faculties of rational cognition as such,

— abstracting, classifying, computing, counting, deducing, identifying patterns

and invariants, operating, ordering, predicting, measuring, schematizing, etc. —

« through [an aprioristic] construction of concepts » (Kant),

that is, independently of the different « ontic domains »

to which they can be applied.

Mathematics *qua* ex-act science

Mathematics is a theatre in which the spirit's acts or operations

— rather than being applied to a particular « ontic » domain » (physics, biology) —

are considered as such, thereby becoming the main characters.

Mathematics is ***ex-act*** in the precise sense that it ***exteriorizes*** or stages

the spirit's ***rational acts*** as such

— e.g. counting, computing, deducing, abstracting, predicting, etc. —

(Cf.: P. Valery reflections on mathematics).

« […] les mathématiques sont un témoignage sur le fonctionnement même de notre esprit […] »

A. Lichnerowiz, *Triangle de pensées*, op.cit., p.39.

The Threefold Doctrine of Science

III. *Theory of the theory of science* or *theory of mathematics*

The very idea of mathematics *qua* science of science

triggers a reflexive structure

such that mathematics itself becomes an object of mathematics.

Category theory is a branch of mathematics

that studies mathematics itself.

Category theory can be understood

as a ***mathematical meta-mathematics***.

Mathematical meta-mathematics *qua* Foundations of Mathematics

« [Category-theoretical] foundations will mean here the study of what is universal in mathematics »

W. Lawvere, *Adjointness in Foundations*, 1969.

Autonomous Foundations

« It would appear that metamathematics is an attempt by mathematicians to lift themselves up by their own bootstraps. »

J. Lambek and P.J. Scott, *Reflections on the Categorical Foundations of Mathematics,* 2011.

Numbers

The fundamental act of rationalization

that grounds the so-called « mathematization of nature »

is the ***quantification*** of natural phenomena,

that is, the assignment of ***quantities*** of different kinds

— tipically, ***numbers*** —

depending on the phenomenon at stake

by ***counting***, ***measuring***, etc.

« Meti […] números nas coisas e era ciência. »

P. Leminski, *Catatau*.

Number as an Operator of Enlightened Disenchantment

« The mythologizing equation of Forms with numbers in Plato's last writings expresses the longing of all demythologizing: **number became enlightenment's canon**. […] Bourgeois society is ruled by equivalence. It makes dissimilar things comparable by reducing them to abstract quantities. **For the Enlightenment, anything which cannot be resolved into numbers** […] **is illusion**. […] All gods and qualities must be destroyed. »

M. Horkheimer and T.W. Adorno, *Dialectics of Enlightenement. Philosophical Fragments*, tr. E. Jephcott, pp.4-5.

« Nunca quis meter um número na minha cachola, queria viver num mundo de qualidade, agora tardio. »

P. Leminski, *Catatau*.

« Nous vivons le temps du despotisme du nombre […] et cependant […] nous ne disposons d'aucune idée récente, active, de ce que c'est qu'un nombre. […] L'impact factuel du nombre n'escorte qu'un silence du concept. […] les nombres […] servent, au sens strict, à tout […] Mais que sont-ils nous l'ignorons […] »

A. Badiou, *Le Nombre et les nombres*, Paris: Éditions du Seuil, 1990, §0.1, p.11.

Rather than trying to save the world

from the “icy waters”

of disenchanted numerical quantification,

we shall try to show that the evolution

of our understanding of the notion of *number*

is an open living adventure of human understanding

— that is, a dehiscence of life itself —

that traverses human histories and geographies.

The thesis according quantification

is concomitante with a disenchantment

presupposes that mathematics

is not itself a living inflorescence of life.

« Ô mathématiques sévères, je ne vous ai pas oubliées, depuis que vos savantes leçons, plus douces que le miel, filtrèrent dans mon cœur comme une onde rafraîchissante. J'aspirais instinctivement, dès le berceau, à boire à votre source, plus ancienne que le soleil, et je continue encore de fouler le parvis sacré de votre temple solennel, moi, le plus fidèle de vos initiés. Il y avait du vague dans mon esprit, un je ne sais quoi épais comme de la fumée ; mais je sus franchir religieusement les degrés qui mènent à votre autel, et vous avez chassé ce voile obscur, comme le vent chasse le damier. Vous avez mis, à la place, une froideur excessive, une prudence consommée et une logique implacable. À votre lait fortifiant, mon intelligence s'est rapidement développée et a pris des proportions immenses, au milieu de cette clarté ravissante dont vous faites présent, avec prodigalité, à ceux qui vous aiment d'un sincère amour. »

Lautréamont, *Chants de Maldoror*.

« Continuo com capacidade de raciocínio – já estudei matemática que é a loucura do raciocínio – mas agora quero o plasma – quero me alimentar diretamente da placenta. »

Clarice Lispector, *Água viva*

« […] la “création” dans mon travail de mathématicien, c’était avant tout là qu’elle se plaçait : dans cette attention intense pour appréhender, dans les replis obscurs, informes et moites d’une chaude et inépuisable matrice nourricière, les premières traces de forme et de contours de ce qui n’était pas né encore et qui semblait m’appeler, pour prendre forme et s’incarner et naître… Dans le travail de découverte, cette attention intense, cette sollicitude ardente sont une force essentielle, tout comme la chaleur du soleil pour l’obscure gestation des semences enfouies dans la terre nourricière, et pour leur humble et miraculeuse éclosion à la lumière du jour. »

A. Grothendieck, *Récoltes et semailles*.

Metaphysical Oppositions

Sublated by the History of the Mathematical Notion of *Quantity*.

- The ***constant*** and the ***variable***.
- The ***finite*** and the ***infinite***.
- The ***discrete*** and the ***continuous***.
- The ***relative*** and the ***universal***.
- The ***local*** and the ***global***.

The central concept we shall consider in this workshop

is the first incarnation of the notion of ***quantity***,

namely the ***constant quantities*** known as ***numbers***.

We shall start with the ***natural numbers***

$$\mathbb{N} = \{0,1,2,3...\}$$

and consider some of its different

finite and transfinite extensions.

Natural Numbers

$$\mathbb{N} = \{0,1,2,3...\}$$

The inclusion of ***zero*** as a number

was a non-trivial conceptual and formal step,

starting in India

(introduced by Brahmagupta in the 7th century BC)

and — after being rejected as a number by the Greeks —

transmitted through the Arabic world

(cf. al-Khwarizmi, 10th century)

into Europe in the 11th century by Fibonacci.

Refs.: R. Kaplan & E. Kaplan, *The Nothing That Is: A Natural History of Zero*, Oxford University Press, 2000; Seife, *Zero: The Biography of a Dangerous Idea*, Penguin Books, 2000.

« God made the integers,

all else is the work of man »

Leopold Kronecker (1823-1891)

Numbers in Greek Mathematics (I)

In Greek mathematics,

numbers are understood exclusively

as ***discrete numbers***

composed of units.

Continuity (i.e. the non-discrete) does not belong to the realm of arithmetics,

but to the realm of ***geometry***.

The opposition between the ***discrete*** and the ***continuous***

grounds the separation between ***arithmetics*** and ***geometry***.

Numbers in Greek Mathematics (II)

Number as a plurality of units in Euclid

« 1. An unit is that by virtue of which each of the things that exist is called one.

2. A number is a multitude composed of units. »

Euclid, *The Elements*, Book VII, Definitions, tr. by T.L. Heath, p.277.

The Discreteness of Being

According to Euclid's definition,

numbers have an ontological scope

given by the fact that anything that exist

is *one thing*.

Numbers in Greek Mathematics (III)

Numbers according to Aristotle

« […] el *ser* de lo uno es *ser* un principio del numero […] lo que es indivisible según la cantidad […] se llama 'unidad' »

Aristoteles, *Metafísica*, 1016b, 15-25, tr. E. Sinnott, Ediciones Colihue, p.349.

« […] todo numero significa una cantidad y […] la unidad […] [es] lo indivisible según la cantidad. »

Aristoteles, *Metafísica*,1089b, 35, *op.cit.*, p.1069.

The Concept of *Natural Number* in the XIX-XX Centuries

- **Logicist definition** (Frege, Russell).

- **Axiomatic or formalist definition** (Peano, Hilbert).

- **Set-theoretic definition** (Dedekind, Cantor, Zermelo, von Neumann Gödel).

The Logicist Definition of *Natural Numbers* (I)

« […] the apparatus of general logical notions […] suffices, without new indefinables or new postulates, to establish the whole theory of cardinal integers as a special branch of Logic. »

B. Russell, *Principles of Mathematics*, 1903, § 107, p.111.

The Logicist Definition of *Natural Numbers* (II)

The logicist definition of natural numbers

exhibit numbers as the ***abstract property***

shared by sets

that can be put in a one-to-one correspondence.

In other terms, natural numbers are defined

as instances of an abstract notion

— called ***cardinal number*** —-

given by the equinumerosity relation.

The Logicist Definition of *Natural Numbers* (III)

Definition: A number associated to a concept *F*

is the set of concepts

whose extension is equinumerous to the extension of *F*.

(Cardinal) numbers are not beings,

but features of (the extensions of) concepts.

They have a logical (rather than ontological) nature.

Ref.: G. Frege, *The Foundations of Arithmetic* [*Die Grundlagen der Arithmetik*], tr. J.L. Austin, Evanston: Northwestern University Press, 1976.

The Logicist Definition of *Natural Numbers* (IV)

Frege's Definition of Numbers

• Given a predicate $P(x)$, we can define a set E_P, the *extension* of $P(x)$, whose elements are the objects a such that $P(a)$ is true.

Note: this presupposes that the set E_P exists for every $P(x)$ (called ***axiom schema of unrestricted comprehension***).

• Two predicates $P(x)$ and $Q(x)$ have the same number if it is possible to establish a bijection between E_P and E_Q (which are then said to be ***equinumerous***).

• The ***number of a predicate*** $P(x)$ is the set of all concepts $Q(x)$ such that their extensions are equinumerous to the extension E_P of $P(x)$.

The Logicist Definition of *Natural Numbers* (V)
The Definition of Zero

For instance, the number 0 is the number

associated to the concept

« being non identical to itself »,

whose extension is empty,

where we are accepting that « there is no entity without identity » (Quine).

« I could have used for the definition of nought any other concept under which no object falls. But I have made a point of one which can be proved to be such on purely logical grounds […] » G. Frege, *The Foundations of Arithmetic*, op.cit., § 74.

Hence, this definition is purely logical

since it depends on a purely logical principle, the identity principle $a = a$.

The Logicist Definition of *Natural Numbers* (V)
The Definition of One

• Frege defines 1 as « the number which belongs to the concept 'identical with 0' », G. Frege, op.cit., § 77, p.90.

• This means that 1 is the set of concepts whose extension is equinumerous to the extension of the concept 'identical with 0'.

• Now, what is the extension of the concept 'identical with 0'?

• Since 0 is the cardinal of the empty set ∅, there is no set other than ∅ that is identical to ∅. Hence, the extension of the concept « Identical to 0 » is given by a set that only contains ∅ (i.e. a « unique » element). There is only one set without elements, namely the empty set ∅.

• All in all, 1 is the collection of all concepts whose extension can be put in a one-to-one correspondence with a set with a « unique » element.

Cf. G. Frege, *The Foundations of Arithmetic*, op.cit., § 77, p.90.

Russell's Paradox
(communicated to Frege in 1903)

Russell exhibits a concept

$P =$ « Being a set that is not an element of itself »

whose extension does not exist (i.e. it is not even empty).

Let's call A the extension of P, i.e. a set of sets. Does A belong to A?

If we assume that $A \in A$, then A contains a set which is an element of itself (which is contradictory with the definition of P).

If we assume that $A \notin A$, then there is a set that is not an element of itself that does not belong to A. Hence, A cannot be the extension of P.

Indeed, it is contradictory to think

that the objects (sets) that fall under this concept form a set.

Such a concept cannot be numbered.

Peano's Axiomatic Definition (I)
(1889)

Peano does not define explicitly

what a natural number is,

but rather specifies a set of axioms

they have to fulfill

(« implicit definition », cf. Schlick, *General Theory of Knowledge*, 1918).

« Toute procedure purement axiomatique introduit des signes *non définis*, dont il n'y aura d'autre presentation dans la pensée que la codification de leur usage par les *axiomes*. »

A. Badiou, *Le Nombre et les nombres*, op.cit., §5.7, p.67.

Peano's Axiomatic Definition (II)
(1889)

- 0 is a natural number.

- Every natural number n has a *successor* $S(n)$:

$1 \doteq S(0)$
$2 \doteq S(1) = S(S(0))$
$3 \doteq S(2) = S(S(1)) = S(S(S(0)))$

- $n \neq n'$ cannot have the same successor (i.e. S is an injection).

- There is no n such that $S(n) = 0$, i.e. 0 is the unique natural number that is not a successor.

- Axiom of Induction (see below).

« I regard the whole of arithmetic as a necessary, or at least natural, consequence of the simplest arithmetic act, that of counting, and counting itself as nothing else than the successive creation of the infinite series of positive integers in which each individual is defined by the one immediately preceding; the simplest act is the passing from an already-formed individual to the consecutive new one to be formed. »

R. Dedekind, *Continuity and Irrational Numbers*, 1872, p.4.

Cantor's *First Principle of Generation*

« The sequence (I) of positive integers 1,2,3…, n, … has its ground of origin [Entstehungsgrund] in the repeated positing [Setzung] and uniting [Vereinigung] of underlying unities [Einheiten], which are regarded as alike; the number n is the expression for a definite finite number [bestimmte endliche Anzahl] of such positings following one another in a sequence; it is also the expression for the unification [Vereinigung] of the posited unities [gesetzten Einheiten] into a whole [zu einem Ganzen]. The formation of the finite real integers [i.e. natural numbers] thus rests upon the principle of adding a unity to an already formed and existing number; I call this principle (which, as we shall soon see, plays an essential role in the generation of the higher integers) the *first principle of generation*. The number [Anzahl] of the numbers n […] formed in this way is infinite and there is no greatest among them. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §11.

Addition of Natural Numbers

- $a + 0 = a$ (i.e. 0 is the neutral element for the operation +).
- $a + S(b) = S(a + b)$

Examples:

$$2 + 1 = 2 + S(0) = S(2 + 0) = S(2)$$

(adding 1 amounts to apply once the successor operator)

$$2 + 2 = 2 + S(1) = S(2 + 1) = S(2 + S(0)) = S(S(2 + 0)) = SS(2) = 4$$

(adding 2 amounts to apply twice the successor operator)

All in all, adding n to a amounts to apply the successor operators to a n times.

« Addition is the combination of any arbitrary repetitions

of the above-mentioned simplest act into a single act »

R. Dedekind, *Continuity and Irrational Numbers,* p.4.

Additive Algebraic Structure of $\mathbb{N}$

It can be shown that the operation $+$ is

- **closed**:

$$a + b \in \mathbb{N} \text{ if } a, b \in \mathbb{N}.$$

- **associative** :

$$(a + b) + c = a + (b + c).$$

- **commutative** :

$$a + b = b + a$$

It follows that $(\mathbb{N}, +, 0)$ is a *commutative monoid,*

where 0 is the neutral element for the addition: $n + 0 = n$.

Note: the operation $+$ is not invertible.

Multiplication of Natural Numbers

- $a \times 0 = 0.$
- $a \times S(b) = a \times b + a$

Examples:

$$2 \times 1 = 2 \times S(0) = 2 \times 0 + 2 = 0 + 2 = 2$$

(the number 1 is the neutral element for the multiplication)

$$2 \times 2 = 2 \times S(1) = 2 \times 1 + 2 = 2 \times S(0) + 2 = 2 \times 0 + 2 + 2 = 4$$

Hence, $n \times n'$ means adding $n + n + \ldots + n$ with n' summands

or $n' + n' + \ldots + n'$ with n summands since $\times$ is commutative).

Multiplicative Algebraic Structure of $\mathbb{N}$

It can be shown that the operation $\times$ is

- **closed**:

$$a \times b \in \mathbb{N} \text{ if } a, b \in \mathbb{N}.$$

- **associative** :

$$(a \times b) \times c = a \times (b \times c).$$

- **commutative** :

$$a \times b = b \times a$$

It follows that $(\mathbb{N}, \times, 1)$ is a *commutative monoid,*

where 1 is the neutral element for the addition: $n \times 1 = n$.

Note: the operation $\times$ is not invertible.

The addition $+$ and the multiplication $\times$ are related by the ***distributivity law***

$$a \times (b + c) = (a \times b) + (a \times c)$$

stating that multiplication $\times$ distributes over addition $+$.

$(\mathbb{N}, +, \times, 0, 1)$ is a ***commutative semi-ring***.

The Axiom of Induction

A predicate $P(n)$ can be demonstrated to be true for every n by

1) showing that $P(0)$ is true.

2) showing that $P(n) \Rightarrow P(n + 1)$ is true.

Example of Mathematical Induction

Prove that

$$\sum_{i=0}^{n} i = 0 + 1 + 2 + \ldots + n = \frac{n(n+1)}{2}$$

Peano and the Infinite

« […] la récurrence permet […] de légiférer dans une totalité infinie *sans parler de son infinité*. Il est clair en effet qu'il y a une infinité de nombres entiers. Parler de 'tous' ces nombres veut donc dire parler d'une infinité actuelle. Mais dans le dispositif axiomatique de Peano, cette infinité n'est pas introduite comme telle. L'axiome de récurrence permet, à partir d'une *vérification* (1 possède la propriété) et d'une *démonstration* implicative (si n possède la propriété, *alors* $n + 1$ la possède aussi) , de conclure que 'tous les nombres possèdent la propriété', sans avoir à s'interroger sur l'extension de ce 'tous'. Le quantificateur universel masque ici la pensée d'un infini actuel : l'infini demeure une forme latente, inscrite au quantificateur sans être délivrée dans la pensée. Peano introduit ainsi le concept de nombre sans transgresser le vieil interdit qui frappe l'infini actuel, interdit qui surplombe encore notre pensée […]. L'axiomatique de Peano évite l'infini, ou la mention explicite de l'infini. »

A. Badiou, *Le Nombre et les nombres*, Paris: Éditions du Seuil, 1990, §5.4, p.64.

Poincaré on Mathematical Induction

« Pourquoi donc [le jugement sur lequel repose le raisonnement par récurrence peut] s'impose-t-il à nous avec une irrésistible évidence ? C'est qu'il n'est que l'affirmation de la puissance de l'esprit qui se sait capable de concevoir la répétition indéfinie d'un même acte dès que cet acte est une fois possible. […] L'induction mathématique, c'est-à-dire la démonstration par récurrence, s'impose […] nécessairement, parce qu'elle n'est que l'affirmation d'une propriété de l'esprit lui-même. »

H. Poincaré, *La science et l'hypothèse*, Ch. I *Sur la nature du raisonnement mathématique.*

« Le sens dont je parle et qui mène l'esprit à se prévoir lui-même, à imaginer l'ensemble de ce qui allait s'imaginer dans le détail, et l'effet de la succession, ainsi résumée, est la condition de toute généralité. »

Paul Valéry, *Introduction à la méthode de Léonard de Vinci*.

General properties of $\mathbb{N}$

• Natural numbers provide the general scheme of ***discreteness***.

« […] 'le nombre" [entier] est apte à saisir la structure des agrégats 'discontinus', ou 'discrets' : les systèmes, souvent finis, formés d"'éléments' ou 'objets' pour ainsi dire isoles les uns par rapport aux autres, sans quelque principe de 'passage continu' de l'un à l'autre », A. Grothendieck, *Récoltes et semailles*.

• The set $\mathbb{N}$ is endowed with a (reflexive, antisymmetric, transitive, and total) relation $\leq$ called a *total order*. Hence they can be used to « measure » a degree of *greatness*.

• According to this order, $\mathbb{N}$ has a smallest element (namely, $0 \in \mathbb{N}$), but not a biggest one. In other terms, every $n \in \mathbb{N}$ has a successor, but not every $n \in \mathbb{N}$ is a successor of another number (being the exception $0 \in \mathbb{N}$, which is not a successor).

• Natural numbers introduce an example of a ***potential infinite***, i.e. of and endless progression.

• They can be used to *count* (cardinals) and *order* (ordinals).

Rather than asking the ontological question

What are numbers?,

we shall ask the pragmatic question

What can we do with (natural) numbers?

Cardinals vs. Ordinales

Cardinal numbers

specify the « size » of a finite set X, its number of elements

(they provide an answer to the question ***How many elements a set X have?***)

Ordinal numbers

specify the position (zeroth, first, second, third…) of an element in a *well-ordered set*

and the *length/order type* of the later:

(they provide an answer to the question ***What is the length of a well-ordered set*** $(X, \leq)$***?***)

[Definition: A *well-ordered* is a set X endowed with total order $\leq$

such that each subset has a minimal element]

« Dans la vision ordinale, le nombre est pensé comme le maillon d'une chaîne, il est élément d'un ordre total. Dans la vision cardinale, il est plutôt la marque d'une 'quantité pure' obtenue par abstraction de domaines d'objets ayant 'même quantité'. Le nombre ordinal est sous le schème d'une *suite*, le nombre cardinal sous celui d'une *mesure*. »

A. Badiou, *Le Nombre et les nombres*, *op.cit.*, p.45.

For finite sets, these notions of cardinal and ordinal number coincide,

that is, the ***length*** or ***order type*** of $(X, <)$

— i.e. the least ordinal that is not a label for an element of the set —

for any order $<$

coincides with the cardinal of the set.

For instance, in the ordered set $X = \{a < b < c\}$,

a is the zeroth element, b the first and c the second

(which means that 3 is the least ordinal number that is not a label for an element of the set).

The ordered set X has length or order type 3 (ordinal) and has 3 elements (cardinal).

The Potential Infinite

The Potential Infinite of $\mathbb{N}$

There is an infinite sequence 0, 1, 2, 3… of natural numbers,

but every natural number *n* is finite.

The term ***potential infinite***

— also called *improper infinite* [*Uneigentlich-Unendliche*] by Cantor —

does not designate a precise number,

but rather the fact that the successor operation *S*

(Cantor's *First Principle of Generation*)

that consists in adding a unity

can be continued forever.

However, this endless process only produces (bigger and bigger) finite numbers.

Potential Infinity According to Aristotle

"For generally the infinite is as follows:

there is always another and another to be taken.

And the thing taken will always be finite, but always different."

Aristotle, *Physics*, 206a27-29.

The « bad infinity » of the infinite progression according to Hegel (I)

« Something becomes an other, but the other is itself a something, hence it likewise becomes an other, and so on and so forth ad infinitum. […] This progression to the infinite is, however, not the true infinite […] [but] the bad infinity of the infinite progression. […] Engaging in the contemplation of this infinite progression is tedious because the same thing is incessantly repeated here. […] there is nothing here but a superficial alternation that remains stuck in the finite. […] If it is then further said that the infinite cannot be reached, then this is quite right, but only because the determination of being something abstractly negative is read into it. Philosophy does not waste its time with such empty and merely transcendent [*Jenseitigen*] things. What philosophy deals with is always something concrete and absolutely present. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline*, §94.

The « bad infinity » of the infinite progression according to Hegel (II)

« I amass colossal numbers,
Millions of mountains,
I pile time upon time
And world upon worlds galore,
And when from this terrifying height
With vertigo I look to you again:
If all the mighty numbers
Were increased thousand-fold,
They would not even be a part of you.

So here we have, first, that constant propelling of quantity and, more specifically, of number beyond itself that Kant describes as horrifying, although the actual horror is perhaps only the boredom of a limit constantly being set and then sublated so that, as a result, one does not make any headway. But, furthermore, the poet mentioned aptly adds to this description of the bad infinity the conclusion:

I remove them, and you lie before me entire

– thereby making it explicit that the true infinite is not to be regarded as something merely beyond the finite and that, to attain consciousness of it, we must renounce that *progressus in infinitum*. »

G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline*, §104.

Aquinas on Infinity

Actual infinite does not exist

because all what exists is (according to the Bible)

ordered « in measure and number »,

but there are no infinite numbers.

« It is written, 'Thou hast ordered all things in measure, and number, and weight' (Wisdom 11:21). [...] every kind of multitude must belong to a species of multitude. Now the species of multitude are to be reckoned by the species of numbers. But no species of number is infinite […] Hence it is impossible for there to be an actually infinite multitude […] But a potentially infinite multitude is possible; because the increase of multitude follows upon the division of magnitude; since the more a thing is divided, the greater number of things result. Hence, as the infinite is to be found potentially in the division of the continuous […] by the same rule, the infinite can be also found potentially in the addition of multitude. »

St. Thomas Aquinas, *Summa Theologica*, Part I, Question 7, Article 4.

Note: Cantor will later make an explicit reference to this paragraph.

Now, what is the obstruction

to the conception

of an actual infinite number?

Equinumerocity via Bijections

Two sets A and B have the same number of elements

(i.e. the same cardinal)

if it is possible to establish

a one-to-one or bijective correspondence

$$f : A \to B$$

between their elements,

i.e. such that for every $y \in B$, there is a unique $x \in A$

such that $f(x) = y$.

Cf.; Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638; Cantor; denied by Bolzano *Paradoxes of the infinite*, 1851, §19-2.

Countable Sets

The cardinality of $\mathbb{N}$

— denoted $|\mathbb{N}|$ —

is called (after Cantor) ***aleph*-0**

and denoted $\aleph_0$.

A set X is *countable*

— i.e. its cardinality is $\aleph_0$, $|X| = \aleph_0$ —

if there is a bijection

$$f : X \to \mathbb{N}$$

i.e. if the elements of X can be put in a one-to-one correspondence with the set $\mathbb{N} = \{0,1,2,3,...\}$.

Galileo's Paradox

Now, it is possible to establish a bijection

$$f : \{0,1,4,9,16,25,...\} \to \mathbb{N}$$

$$n^2 \mapsto n$$

Hence, there are as many natural numbers

0,1,2,3,4,...

as square numbers

0,1,4,9,16,...

Therefore, a strict part (the set of square numbers)

is as big as the whole ($\mathbb{N}$),

which contradicts Euclid's Common Notion 5 « The whole is greater than the part ».

Galileo on Infinity (I)

• **Simplicio**: Here a difficulty presents itself which appears to me insoluble. Since it is clear that we may have one line greater than another, each containing an infinite number of points, we are forced to admit that, within one and the same class, we may have something greater than infinity, because the infinity of points in the long line is greater than the infinity of points in the short line. This assigning to an infinite quantity a value greater than infinity is quite beyond my comprehension.

• **Salviati**: This is one of the difficulties which arise when we attempt, with our finite minds, to discuss the infinite, assigning to it those properties which we give to the finite and limited; but this I think is wrong, for we cannot speak of infinite quantities as being the one greater or less than or equal to another. To prove this I have in mind an argument which, for the sake of clearness, I shall put in the form of questions to Simplicio who raised this difficulty. I take it for granted that you know which of the numbers are squares and which are not.

Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638.

Galileo on Infinity (II)

• **Simplicio**: I am quite aware that a squared number is one which results from the multiplication of another number by itself; thus 4, 9, etc., are squared numbers which come from multiplying 2, 3, etc., by themselves.

• **Salviati**: Very well; and you also know that just as the products are called squares so the factors are called sides or roots; while on the other hand those numbers which do not consist of two equal factors are not squares. Therefore if I assert that all numbers, including both squares and non-squares, are more than the squares alone, I shall speak the truth, shall I not?

• **Simplicio**: Most certainly.

• **Salviati**: If I should ask further how many squares there are one might reply truly that there are as many as the corresponding number of roots, since every square has its own root and every root its own square, while no square has more than one root and no root more than one square.

• **Simplicio**: Precisely so.

Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638.

Galileo on Infinity (III)

• **Salviati**: But if I inquire how many roots there are, it cannot be denied that there are as many as the numbers because every number is the root of some square. This being granted, we must say that there are as many squares as there are numbers because they are just as numerous as their roots, and all the numbers are roots. Yet at the outset we said that there are many more numbers than squares, since the larger portion of them are not squares. Not only so, but the proportionate number of squares diminishes as we pass to larger numbers, Thus up to 100 we have 10 squares, that is, the squares constitute 1/10 part of all the numbers; up to 10000, we find only 1/100 part to be squares; and up to a million only 1/1000 part; on the other hand in an infinite number, if one could conceive of such a thing, he would be forced to admit that there are as many squares as there are numbers taken all together.

• **Sagredo**: What then must one conclude under these circumstances?

• **Salviati**: So far as I see we can only infer that the totality of all numbers is infinite, that the number of squares is infinite, and that the number of their roots is infinite; neither is the number of squares less than the totality of all the numbers, nor the latter greater than the former; and finally *the attributes "equal," "greater," and "less," are not applicable to infinite, but only to finite, quantities*. When therefore Simplicio introduces several lines of different lengths and asks me how it is possible that the longer ones do not contain more points than the shorter, I answer him that one line does not contain more or less or just as many points as another, but that each line contains an infinite number.

Galileo, *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, 1638.

The Infinite Dilemma

Either

♣ we deny the concept of an infinite number and the possibility of an actual infinity (as Aristotle, Aquinas, Galileo and Leibniz did);

♥ or we limit the validity of Euclid's Common Notion 5 « The whole is greater than the part » to finite numbers (as Cantor did).

The Cartesian *Indefinite* (I)

Descartes warns us

that we cannot infer the existence of an actual infinite

(a notion that can only be applied to God)

from the mere absence of limits.

« […] the Cardinal of Cusa and other theologians have supposed the world to be infinite […] my opinion is easier to accept than theirs, because I say not that the world is infinite but only that it is *indefinite*. There's a quite notable difference between the two: because if we say that something is infinite we should have a reason that tells us that it is so; and we can't have such a reason except in the case of God; but to say that a thing is indefinite, all we need is *not* to have a reason showing that it has bounds. »

R. Descartes, *Letter to Chanut*, June 6th, 1647.

The Cartesian *Indefinite* (I)

« [...] we must never discuss the infinite, but must simply consider those things in which we notice no limits as indefinite; as, for instance, the extension of the world, the divisibility of parts of matter, the number of stars, etc. Thus we shall never be wearied by any debates concerning the infinite. [...] inasmuch as we are finite, it would be absurd for us to attempt to determine anything concerning the infinite [...] surely only those who judge their own mind to be infinite ought to think about such things. Moreover, we shall not affirm that all those things in which we have been able to find no limit after some consideration are infinite; but shall view them as indefinite. [...] And we shall say that these things [the extension of the world, the divisibility of parts of matter, the number of stars] are indefinite rather than infinite: both in order to reserve the term ‘infinite' for God alone; because in Him alone, in every respect, we not only recognize no limits [to His perfection], but also in a positive sense understand that there are none […] »

R. Descartes, *Principles of Philosophy*, § 26 and 27, 1644.

Leibniz on the Infinite

I. The Infinite is not a Number

« I concede the infinite plurality of terms, but this plurality itself does not constitute a number or a single whole. It means […] that there are more terms than can be designated by a number. Just so there is a plurality or a complex of all numbers, but this plurality is not a number or a single whole. »

G.W. Leibniz, Letter to Bernoulli, February 21, 1699.

Leibniz on the Infinite

II. The Infinite *qua* Number is a Useful Fiction

« [...] the infinite is not a whole, but only a fiction, since otherwise the part would be equal to the whole. » (Leibniz, quoted in Arthur & Rabouin, 2025, p.172).

« I no more support infinitely small magnitudes than infinitely large ones, or no more infinitesimals than infinituples. For I consider both to be fictions of the mind, due to abbreviated ways of speaking, which are suitable for calculation, in the way that imaginary roots in algebra are. »

G.W. Leibniz, Letter to Des Bosses, March 11, 1706.

« I also deny that properly speaking there exists an infinite or an infinitely small number, or an infinite or infinitely small line—even though Euclid often speaks of an infinite line, but in a sound sense. The *infinite*, whether *continuous* or *discrete*, is properly neither one, nor a whole, nor a quantity [*quantum*]; and if we take it as such by a certain analogy, it is only, so to speak, a way of speaking; when namely, there are more things than can be comprised by any number, yet by analogy we attribute to these things a number, which we call infinite. »

G.W. Leibniz, An Observation That Ratios or Proportions Do Not Hold for Quantities Less Than Zero, and on the True Meaning of the Infinitesimal Method, quoted in R.T.W. Arthur & D. Rabouin, *Leibniz on the Foundation of the Differential Calculus*, p.245.

Leibniz on the Infinite

III. The Existence of the Actual Infinite

« I am so in favor of the actual infinite that instead of admitting that Nature abhors it, as is commonly said, I hold that Nature makes frequent use of it everywhere, in order to show more effectively the perfections of its Author. Thus I believe that there is no part of matter which is not, I do not say divisible, but actually divided; and consequently the least particle ought to be considered as a world full of an infinity of different creatures. »

G.W. Leibniz, Letter to Foucher, January 1692.

Leibniz on the Actual Infinite of Minute Perceptions

« […] at every moment there is in us an infinity of perceptions, unaccompanied by awareness or reflection; that is, of alterations in the soul itself, of which we are unaware because these impressions are either too minute and too numerous, or else too unvarying, so that they are not sufficiently distinctive on their own. But when they are combined with others they do nevertheless have their effect and make themselves felt, at least confusedly, within the whole. […] To give a clearer idea of these minute perceptions which we are unable to pick out from the crowd, I like to use the example of the roaring noise of the sea which impresses itself on us when we are standing on the shore. To hear this noise as we do, we must hear the parts which make up this whole, that is the noise of each wave, although each of these little noises makes itself known only when combined confusedly with all the others, and would not be noticed if the wave which made it were by itself. We must be affected slightly by the motion of this wave, and have some perception of each of these noises, however faint they may be; otherwise there would be no perception of a hundred thousand waves, since a hundred thousand nothings cannot make something. […] These minute perceptions […] are […] vivid in the aggregate but confused as to the parts; those impressions which are made on us by the bodies around us and which involve the infinite; that connection that each being has with all the rest of the universe. […] noticeable perceptions arise by degrees from ones which are too minute to be noticed. To think otherwise is to be ignorant of the immeasurable fineness of things, which always and everywhere involves an actual infinity. »

G.W. Leibniz, *New Essays on Human Understanding*, Preface, 53-7.

Deleuze on Leibniz's Theory of Minute Perceptions

« Et ce sont ces petites perceptions obscures, confuses, qui composent nos macroperceptions, nos aperceptions conscientes, claires et distinctes : jamais une perception consciente n'arriverait si elle n'intégrait un ensemble infini de petites perceptions […] Les petites perceptions […] sont […] des 'différentiels de la conscience'. […] Pour Maïmon comme pour Leibniz, la détermination réciproque des différentielles ne renvoie pas à un entendement divin, mais aux petites perceptions comme représentants du monde dans le moi fini […] L'infini actuel dans le moi fini […] Les petites perceptions constituent l'obscure poussière du monde inclus dans chaque monade, le sombre fond. Ce sont les rapports différentiels entre ces actuels infiniment petits qui *tirent au clair*, c'est-à-dire qui constituent une perception claire […] Toujours les rapports différentiels sélectionnent les petites perceptions qui entrent dans chaque cas, et produisent ou tirent la perception consciente qui en sort. Aussi le calcul différentiel est-il le mécanisme psychique de la perception […] *Je produis un accord* chaque fois que je peux établir dans un ensemble d'infiniment petits des rapports différentiels qui rendront possible une intégration de l'ensemble, c'est-à-dire une perception claire et distinguée. C'est un filtre, une sélection. »

G. Deleuze, *Le pli. Leibniz et le baroque*, pp.115, 118-9 & 178.

Gauss against actual infinity

« I protest against the use of an infinite quantity as an actual entity; this is never allowed in mathematics. The infinite is only a manner of speaking, in which one properly speaks of limits to which certain ratios can come as near as desired, while others are permitted to increase without bound. »

C.F. Gauss, Letter to Heinrich Schumacher, July 12, 1831.

The *horror infiniti* or *apeirophobia*

« If, however, out of a justified aversion against such an *illegitimate* actual infinite, a certain *Horror Infiniti,* which found its classic expression and support in the mentioned letter of Gauss, has been formed in broad layers of science, under the influence of the modern Epicurean-materialistic tendency of our time, so the therewith-connected uncritical rejection of the *legitimate* actual infinite seems to me to be no trifling offense against the nature of things, which one has to take as they are, and this behavior can be understood as a kind of *shortsightedness,* which deprives one of the possibility to see the actual infinite, although it in its Supreme, Absolute Bearer has created us and preserves us, and in its secondary, transfinite forms surrounds us everywhere [allüberall] and even dwells in our mind. »

G. Cantor, *On the Various Standpoints With Regard to the Actual Infinite*, letter to Mr. G. Eneström.

The Infinite and the Sublime in Kant (I)

In Kant (*Critique of Judgement*, 1790), the sublime is the « absolutely great »

- beyond comparison;
- beyond all what can be grasped by the senses (that is, it is the suprasensible);
- beyond numerical quantification;
- beyond the progress *ad infinitum* (that is, it is an « absolute totality »).

The Infinite and the Sublime in Kant (II)

« Sublime is the name given to what is absolutely great. […] to assert […] that something is great, is quite a different thing from saying that it is absolutely great (*absolute, non comparative magnum*). The latter is what is beyond all comparison great. […] If […] we call anything not merely great, but […] absolutely […] great, that is to say, sublime, we soon perceive that for this it is not permissible to seek an appropriate standard outside itself, but merely in itself. It is a greatness comparable to itself alone. Hence it comes that the sublime is not to be looked for in the things of nature, but only in our own ideas. […] Nothing […] which can be an object of the senses is to be termed sublime […]. But precisely because there is a striving in our imagination towards progress *ad infinitum*, while reason demands absolute totality […], is the awakening of a feeling of a supersensible faculty within us […] *The sublime is that, the mere capacity of thinking which evidences a faculty of mind transcending every standard of the senses.* »

I. Kant, *Critique of judgment*, 1790, §26.

The Infinite and the Sublime in Kant (III)

« But the infinite is absolutely (not merely comparatively) great. In comparison with this all else (in the way of magnitudes of the same order) is small. But the point of capital importance is that the mere ability even to think it as a whole indicates a faculty of mind transcending every standard of the senses. For the latter would entail a comprehension yielding as unit a standard bearing to the infinite a definite ratio expressible in numbers, which is impossible. Still the mere ability even to think the given infinite without contradiction, is something that requires the presence in the human mind of a faculty that is itself supersensible. »

I. Kant, *Critique of judgment*, 1790, §26.

Hilbert on Infinite

« Just as in the limit processes of the infinitesimal calculus, the infinite in the sense of the infinitely large and the infinitely small proved to be merely a figure of speech, so too we must realize that the infinite in the sense of an infinite totality […] is an illusion. […] the infinite is nowhere to be found in reality, no matter what experiences, observations, and knowledge are appealed to. […] It neither exists in nature nor provides a legitimate basis for rational thought — a remarkable harmony between being and thought. […] The role that remains for the infinite to play is solely that of an idea — if one means by an idea, in Kant's terminology, a concept of reason which transcends all experience and which completes the concrete as a totality. »

D. Hilbert, *Über das Unendliche*, 1925.

Towards the Actual Infinite

The Leap Into the Beyond

« The leap into the beyond occurs when the sequence of numbers that is never complete but remains open toward the infinite is made into a closed aggregate of objects existing in themselves. Giving the numbers the status of objects becomes dangerous only when this is done. The belief in the absolute is deeply implanted in our breast; no wonder, then, that mathematics was bold and naive enough to perform the leap. [...] the system of numbers has become a realm of absolute existence which is 'not of this world' and from which only gleams here and there are caught and reflected in our consciousness. »

H. Weyl, *Philosophy of Mathematics and Natural Science.*

Cantor on Aquinas

« The thoughtful arguments which one finds here and with the church fathers (Augustine should be excepted) *against* the standpoint represented by me were thoroughly justified, and it was therefore right and correct to oppose actually infinite numbers, so long as a principle of *individuation*, *specification* and *ordination* of that actual infinite which I call the *transfinite* had not been found. However, all these arguments crumble as soon as one set up such a principle and it had been demonstrated as *true*. This however one finds in my works. »

G. Cantor, Nachlass Vl , p. 110.

Cantor's determinate and complete transfinite numbers

« I have been logically compelled [...] to the point of view which considers the infinitely great not merely in the form of something growing without limits [unbegrentz Wachseden] [...] but also to fix it mathematically by numbers in the determinate form of the completed infinite [Vollendet-unendlichen] following the finite there is a *Transfinitum* [...] that is, there is an unbounded step-ladder of determinate modes which in their nature are not finite but infinite, but which just like the finite, can be determined by well-defined and distinguishable *numbers*. I am convinced that the domain of definable quantities is *not* exhausted by the finite quantities, and the bounds of our knowledge may accordingly be extended without violence to our nature. [...] All things, finite of infinite are *definite* and, God excepted, can be determined by the intellect. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §4.

How can we label with ordinal numbers

the set $\mathbb{N}$ when we endow it

with the following order:

0,2,4,...,1,3,5,...?

Cantor's answer

We shall introduce a new number ω

which is the ordinal number

corresponding to the position of the term 1

in the well-ordered set

0,2,4,...,1,3,5,...

« L’idée surgit alors (ou le désir) de précipiter le cours de cette suite, d’en porter les termes à leur limite, à celle de leurs expressions imaginables, après laquelle tout sera changé. »

Paul Valéry, *Introduction à la méthode de Léonard de Vinci*.

Cantor called ω a ***limit ordinal***

in the sense that it is the limit number

towards which the series 0,1,2,3,... tends.

The successor operator S allows us

to infinitely tend towards the infinite number ω

without ever reaching it,

which means that ω cannot be generated

by applying Cantor's first principle of generation.

ω is the length or order type of the well-ordered set

$0 < 1 < 2 < 3 < \ldots$

(i.e. the first ordinal that does not label an element of the set).

Like 0,

the ordinal ω is not a successor,

i.e. it cannot be reached from a previous number

by applying the successor operation S.

However, ω has a successor, namely $\omega + 1$, even if it is transfinite.

Transfinite ordinals might not have a predecessor (e.g. ω),

but they always have a successor.

Cantor's First Transfinite Ordinal

« The number [Anzahl] of the numbers n [1,2,3,...] of class (I) formed in this way is infinite and there is no greatest among them. However contradictory it might be to speak of a greatest number of class (I), there is nevertheless nothing offensive in thinking of a new number which we shall call ω, and which will be the expression for the idea [fact] that the entire assemblage [aggregate, Inbegriff] (I) is given in its natural, orderly succession [natural succession according to a law] (Similarly n is an expression for the fact that a certain finite number [Anzahl] of unities is united to form a whole [zu einem Ganzen vereinigt wird].) We may even imagine the newly created number ω as a limit which the numbers n approach, provided that we understand nothing more thereby than that ω is the first integer which follows all numbers n. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §11.

Limit Ordinals According to Mallarmé

A limit ordinal is a new position (*un endroit*)

beyond (*au-délà*) all finite positions.

« Aussi loin qu'un endroit fusionne avec un au-delà »

S. Mallarmé.

Cantor's *Second Principle of Generation* of natural numbers

« The logical function which gave us the two numbers ω and 2ω is obviously different from the *first* principle of generation. I call it the *second principle of generation* of real whole numbers and define them more precisely: if any definite succession of defined whole real numbers exists, for which there is no largest, then a new number is created by means of this second principle of generation which is thought of as the *limit* of those numbers, i.e., it is defined as the next number larger than all of them. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §11.

The notion of ***order type*** of an infinite well-ordered set

(i.e. the first ordinal number that does not label an element in the set)

is order-dependent.

For instance, the order type of

0,1,2,3,...

is ω

whereas the the order type of

0,2,4,...,1,3,5,...

is $\omega \cdot 2$.

In the realm of transfinite numbers,

the notions of *cardinal number*

and *ordinal number*

do not coincide.

For instance, 0,1,2,3,... and 0,2,4,...,1,3,5,...

have the same cardinal $\aleph_0$

(where a bijection is given by $n \mapsto n$)

but different ordinals (ω and $\omega \cdot 2$ respectively).

In the realm of transfinite numbers,

ordering a set *X*

— i.e. labeling the elements *X* with the ordinals *zeroth*, *first*, *second*,…

and assigning the corresponding length/order type to *X* (*Anzahl* in Cantor's terms) —

is not the same as ***counting*** the number of elements of *X* (« power » [*Mächtigkeit*] in Cantor's terms).

The "number of elements" of such a set

is given by the cardinality

— or « power » [*Mächtigkeit*] in Cantor's terms —

of the resulting ordinal.

« It is well known that in the Middle Ages, the '*infinitum actu non datur*' (the actual-infinite does not exist) was always represented by all scholastics as an irrefutable sentence taken from Aristotle. If, however, one considers the reasons which Aristotle gives against the real existence of the infinite (see for example his *Metaphysics*, Book XI, Chapter 10), then they essentially lead back to one presupposition, involving one *petitio principii* (begging the question), based on the assumption that there are only finite numbers, which he concluded from the fact that he was only aware of things in finite quantities. I believe, however, to have proved above, and this will be shown even more clearly in the remainder of this work, that definite countings can be made on infinite sets as well as on finite sets, provided that the sets follow a certain rule which is that they can be well-ordered sets. Without such a rule regarding the ordering of the elements of a set no counting can be done on the set — this is in the nature of the concept of counting. Even in the case of finite quantities, a counting can only be carried out for a certain sequence of the counted elements, but here it shows as a special nature of finite quantities that the result of the counting — the *ordinal-number* [*Anzahl*]— is independent of the respective arrangement; while in the case of infinite quantities, as we have seen, such independence in general does not apply, but the number of an infinite set is an infinite integer which is *also determined* by the rule of counting. This is where the essential difference between the finite and the infinite lies, which is based upon nature itself and can therefore never be eradicated, but the existence of the infinite will never henceforth be denied on account of this difference […] »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §4.9.

A part can have the same cardinal as the whole

« Let M be the totality (n) of all finite numbers n, M' the totality ($2n$) of all even numbers $2n$. Here it is definitely correct to say that according to its entities M is richer than M'; indeed, M contains in addition to the even numbers, which make up M, also all the uneven numbers M''. On the other hand, it is also definitely correct that both sets M and M' […] have the same cardinal number. Both (propositions) are certain and they do not conflict with each other if one carefully observes the distinction between reality and number. One should therefore say: the set M has more reality than M', because it contains as parts M' and M'' in addition; the cardinal numbers corresponding to them are however equal. When will these easy and enlightening truths be finally acknowledged by all thinkers?

G. Cantor, *Mitteilungen zur Lehre vom Transfiniten* (quoted in P. Mancosu, *Abstraction and Infinity*, p.132).

Hilbert's Hotel (I)

In a hotel with a countably infinite number of rooms,

the fact that

all rooms are occupied

does not imply that

there is no more space for new guests.

Hilbert's Hotel (II)

« Let us take the simplest example: the set of integers. Here the statement "The part is smaller than the whole" no longer holds. This important fact can easily be illustrated by means of the example of the occupied hotel.

We now assume that the hotel has infinitely many rooms numbered 0,1,2,3,4,5,… and that each of the rooms is occupied by a single guest. All that the manager has to do in order to accommodate a new guest is to make sure that each of the old guests moves to a new room with the number one unit larger. In this way room 0 becomes available for the new guest. [I.e. the application $n \mapsto n+1$ between the sets {0,1,2,3,...} and {1,2,3,...} is a bijection, which means that a part (the set {1,2,3,...}) has as many elements as the whole {0,1,2,3,...}] One can of course make room for any finite number of new guests in the same manner; and thus, in a world with an infinite number of houses and occupants there will be no homeless.

[…] It is even possible to get space for an infinite number of new guests […]. One could, for example, ask the old guest who originally occupied room number n to move to room number $2n$. In this way infinitely many rooms with odd numbers would be left free for new guests. [I.e. the application $n \mapsto 2n$ between the sets {0,1,2,3,...} and {2,4,6,...} is a bijection, which means that a part (the set of even numbers {2,4,6,...}) has as many elements as the whole set of natural numbers {0,1,2,3,...}]»

D. Hilbert, *David Hilbert's Lectures on the Foundations of Arithmetics and Logic 1917-1933*, ed. by W. Ewald and W. Sieg, 2013, p.730.

Hilbert's Hotel (III)

Hence, all the guests in the « bigger » set of rooms {0,1,2,3,...}

can be accommodated in the « smaller » set of rooms {1,2,3,...}

through the bijective map $n \mapsto n + 1$,

thereby liberating the room with the label 0.

(Re)learning to Count (I)

$$0,1,2,3,...$$

$$0,1,2,3,...,\omega$$

$$0,1,2,3,...,\omega,\omega+1,\omega+2,...,\omega+\omega=\omega\cdot 2$$

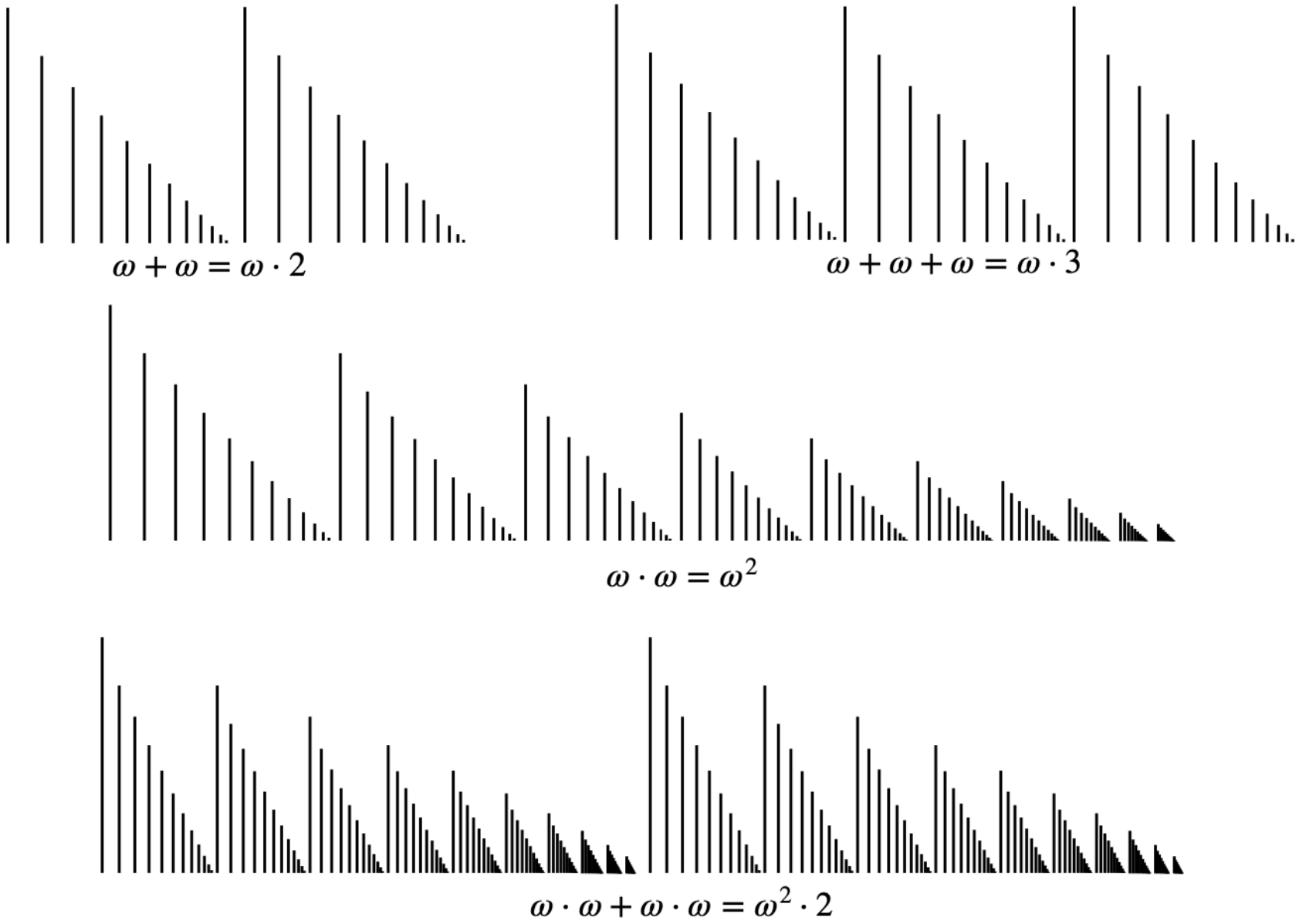
$$\omega\cdot 2+1,\omega\cdot 2+2,...,\omega\cdot 2+\omega=\omega\cdot 3$$

$$\omega,\ldots,\omega\cdot 2,\omega\cdot 3,...,\omega\cdot\omega=\omega^2$$

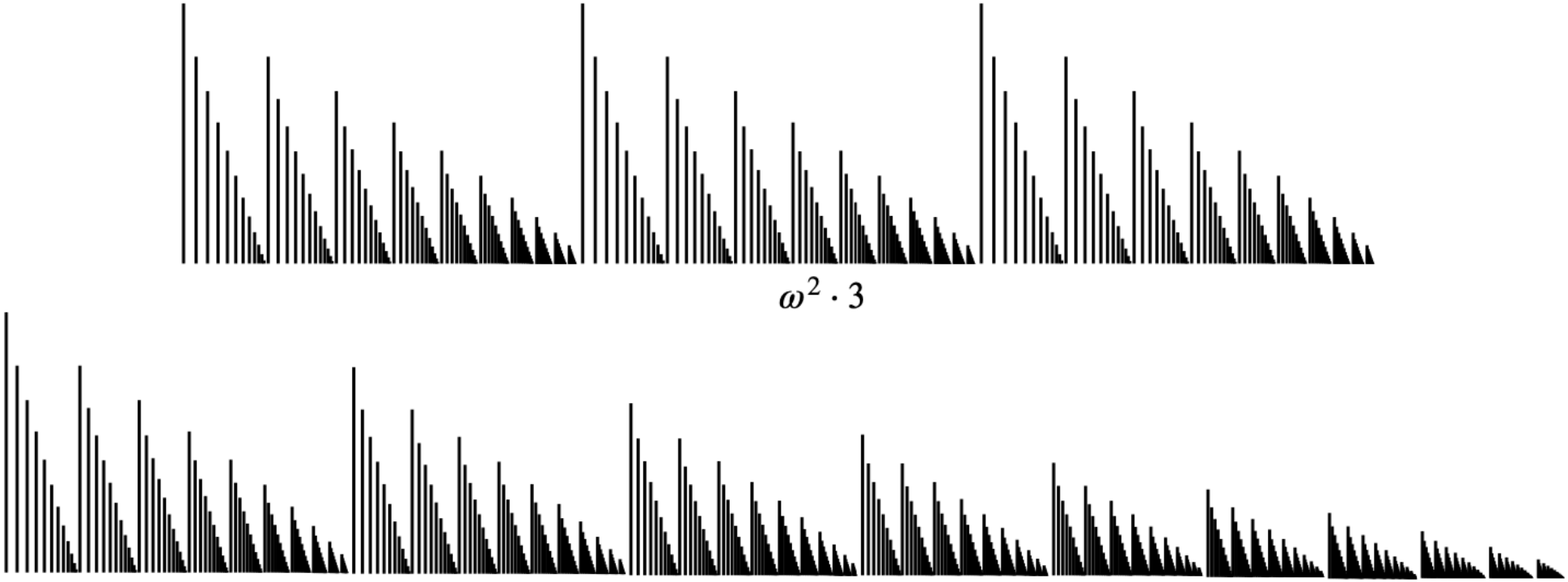
For instance, ω^2 is the order type of an encyclopedia

with countably infinite volumes,

each of which has countably infinite pages.



Ref. K. Easwaran; A. Hájek; P. Mancosu; G. Oppy, Infinity, *The Stanford Encyclopedia of Philosophy (Spring 2022 Edition)*, Edward N. Zalta (ed.).



Ref. K. Easwaran; A. Hájek; P. Mancosu; G. Oppy, Infinity, *The Stanford Encyclopedia of Philosophy (Spring 2022 Edition)*, Edward N. Zalta (ed.).

ω^3 is the order type

of a countably infinite number of encyclopedias,

each with countably infinite volumes,

each with countably infinite pages.

(Re)learning to Count (II)

Starting to count from ω^2, we obtain the following ordinals:

$$\omega^2 + 1,...,\omega^2 + \omega, \omega^2 + \omega + 1,...,\omega^2 + \omega \cdot 2,...,\omega^2 \cdot 2,...,\omega^3, \ldots, \omega^4, \ldots \omega^\omega$$

$$\omega^\omega + 1,...,\omega^\omega + \omega^\omega = 2\omega^\omega, \ldots, \omega\omega^\omega = \omega^{\omega+1}, \ldots, \omega^{2\omega}, \ldots, \omega^{\omega^2}, \ldots \omega^{\omega^\omega}$$

$$\epsilon_0 \equiv \omega^{\omega^{\omega^{\cdots}}}$$

(Re)learning to Count (III)

Starting to count from

$$\epsilon_0 \equiv \omega^{\omega^{\omega^{\cdots}}},$$

we obtain the ordinals:

$$\epsilon_1 \equiv (\epsilon_0)^{(\epsilon_0)^{(\epsilon_0)^{\cdots}}},$$

$$\epsilon_2 \equiv (\epsilon_1)^{(\epsilon_1)^{(\epsilon_1)^{\cdots}}}, \ldots, \epsilon_\omega, \epsilon_{\omega+1}, \ldots, \epsilon_{\epsilon_0}, \ldots, \epsilon_{\epsilon_1}, \ldots, \epsilon_{\epsilon_\omega}, \ldots, \epsilon_{\epsilon_{\epsilon_0}}, \ldots, \epsilon_{\epsilon_{\epsilon_1}}, \ldots, \epsilon_{\epsilon_{\epsilon_\omega}}, \ldots \epsilon_{\epsilon_{\epsilon_{\epsilon_0}}}, \ldots,$$

$$\eta_0 = \epsilon_{\epsilon_{\epsilon_{\epsilon_{\cdots}}}}$$

Cantor on the Finite, the Infinite and the Absolute (I)

« As different as the theories of these writers [Locke, Descartes, Spinoza, Leibniz] are, in their judgment of the finite and the infinite they essentially agree that the concept of a number includes the finiteness of it, and that on the other hand the true infinite or absolute, which resides in God, is not permitted any determination. As far as the latter point is concerned, I fully agree with it, as it cannot be otherwise, because the sentence: *"omnis determinatio est negatio"* (every determination is negative) is, for me, completely beyond question. On the other hand […] I see in the former a *petitio principii* (begging the question), which renders explainable many contradictions which can be found in all these authors and especially in Spinoza and Leibniz. The assumption is that, apart from the finite, and the absolute infinite that cannot be reached by any determination, there can be no new entities which are infinite but at the same time are determinate numbers, the ones that I refer to as the actual-infinite. I do not find that this assumption is justified by anything and, in my opinion, it even stands in contradiction to certain assertions put forward by the two aforementioned philosophers. What I maintain and believe to have proved […] is that after the finite there is a *transfinite* (which might also be called a *suprafinite*)—that is, there is an unlimited scale of definitive steps, which in their nature are not finite but rather infinite, but which, like finite entities, can be determined by definite, well-defined and mutually distinguishable numbers. My conviction is therefore that the finite magnitudes do not constrict the realm of definable magnitudes and that the limits of our knowledge can be extended accordingly without our nature having to apply any limitation. Instead of the Aristotelian school sentence discussed above in § 4, I therefore put another:

'*Omnia seu finita seu infinita definita sunt et excepto Deo ab intellectu determinari possunt.*'

(All things finite or infinite are *definite* and, except for God, can be determined by the intellect). »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §5.3.

Cantor on the Finite, the Infinite and the Absolute (II)

« [The Absolute is the veritable infinity whose magnitude is such that it] cannot in any way be added to or diminished, and it is therefore to be looked upon quantitatively as an absolute maximum. In a certain sense it transcends the human power of comprehension, and in particular is beyond mathematical determination. »

G. Cantor, quoted in M. Hallett, *Cantorian set theory and limitation of size*, p.13.

Cantor on the Finite, the Infinite and the Absolute (III)

« The finiteness of human understanding is very often cited as the reason why only finite numbers can be the subject of thought, but I see in this assertion again the aforementioned circular argument. […] The words 'finiteness of understanding', which one hears so often, do not, I believe, in any way apply; as limited as human nature is, in truth it is much attached to the infinite and I believe that even if it were not itself infinite in many aspects, we all have a firm confidence and certainty regarding the being of the absolute, and in which we all agree is known to be unexplainable. And in particular I take the view that the human mind has an unlimited capacity for the stepwise formation of sets of integers which are related to the infinite steps and whose *cardinal-numbers* are of increasing magnitude. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §5.4.

Cantor on the Finite, the Infinite and the Absolute (IV)

« There is no doubt for me that we will continue on this path, never reaching an insurmountable boundary, but also not reaching any sort of comprehension of the absolute. The absolute can only be recognized, but never known, not even approximately known. […] As Albrecht von Haller says of eternity: 'I attain to the enormous number, but you, O eternity, lies always ahead of me.' »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, Endnotes 2.

Cantor on the Essence of Mathematics as Freedom

« [mathematics] *only* takes into account the *fundamental* reality of its concepts in the formation of its ideas and is therefore *not* duty-bound to also examine them according to their *ephemeral* reality. Because of this exceptional standpoint, which distinguishes it from all other sciences […], it particularly deserves the name of *free mathematics* […].

Mathematics is completely free in its development and is only bound to the self-evident consideration that its concepts are both free of contradictions and that they are in fixed relationships to proven concepts that have already been previously established. In particular, […] [o]nce a number satisfies all these conditions, it can and must be considered as existing and real in mathematics. Regarding this one can now see, […] the reason why one has to consider that the rational, irrational and complex numbers exist, in exactly the same way as we consider the finite positive integers to exist.

[…] the *essence of mathematics* lies precisely in its *freedom*. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical Investigation Into the Theory of the Infinite* [Grundlagen], 1883, §8.3-5.

Summary of Cantor's Transfinite Revolution

Pre-Cantorian Conception of the Infinite

- There is no actual infinite, only potential infinite, i.e. the infinite of an endless progression (Aristotle).

- The infinite is « beyond measure » or determination, i.e. it cannot be measured or determined by a number (Aquinas, Galileo, Leibniz).

- It is unique, *the* infinite.

« […] what is infinite is not comparable as a greater or smaller; there cannot therefore be a relation of infinite to infinite according to orders or dignities of the infinite […] »

G.W.F. Hegel, *The Science of Logic*, p. 216.

Cantorian Conception of the Infinite

- There exists many actual infinities associated with different well-determined transfinite numbers.

- In the realm of infinite numbers, certain properties of finite numbers do not hold:

  .Coincidence between cardinals and ordinals.

  .« The whole is greater than the part » (Euclid's Common Notion 5).

- The classical dichotomy between *finite understanding* and the *divine infinite* becomes complexified into a triptych: the *Finite*, the *Transfinite*, and the *Absolute,* where the first one and the second one admit numerical determination and are fully accessible to human understanding.

Finite extensions of $\mathbb{N}$

Up to know, we have enriched $\mathbb{N}$

by adding transfinite numbers bigger than any n.

We shall now explore extensions of $\mathbb{N}$

obtained by adding

finite numbers.

Why should we

introduce new numbers?

The « Speculative » or « Dialectical » Dynamics of Reason

Given a number system A,

there are « problems » formulated

by using the numbers in A

that do not have « solutions » in A.

These « problems » are like symptoms

that inscribe within A its potential extensions.

What are the solutions of the $\mathbb{N}$-equation

$$x + 1 = 0?$$

An equation formulated using numbers in $\mathbb{N}$

does not have a solution in $\mathbb{N}$

or, equivalently,

the result of an operation

between numbers in $\mathbb{N}$ is not in $\mathbb{N}$.

« What we cannot speak about

we must pass over in silence. »

L. Wittgenstein, *Tractatus Logico-Philosophicus*, §7.

Historical and Dialectical Mathematics

What we cannot speak about

we must expand our « transcendental »

(linguistic, numerical) resources…

… in order to avoid falling prey

under the « precritical syndrom »,

which consists in mistaking a limitation of our « transcendental resources »

for an insight into necessity

(D.C. Dennett, *Consciousness Explained*).

Lacan against Wittgenstein

« Lacan's formula, according to which whatever is excluded from the symbolic reappears in the real, can be interpreted here as follows: in certain conditions, the excluded proper to an *already produced* mathematical structure reappears as the instigating mark of a real (historical) process of *production* of a different structure. »

A. Badiou, *Infinitesimal Subversion*.

The Maximalist Ethos of the Mathematician

We shall always introduce

all the « ideal elements » we need

in order

- to guarantee the closure of a given operational domain;
- to find solutions to a given equation (history of number systems);
- to avoid exceptions (points at infinity in projective geometry).

The new « ideal elements » were successively described as

fictitious *(« fictions of the mind »)*,

ideal (i.e. non real),

irrational,

imaginary,

ghostly elements

introduced to fulfill a desire for completeness, elegance or symmetry.

Invariable, these pure « inventions » of the mind

firstly introduced as mere « symbols » obeying certain rules

gained « reality » through systematic use.

« Casi inmediatamente, la realidad cedió en más de un punto. Lo cierto es que anhelaba ceder. »

J.L. Borges, *Tlön, Uqbar, Orbis Tertius*.

« The expression ‘ideal elements’ is only justified from the point of view of the original system. In the new system, we no longer distinguish between real and ideal elements. »

D. Hilbert, *Die Rolle von idealen Gebilden*.

« […] the sharp distinction between the intuitive [Anschaulichen] and the non-intuitive [Nicht-Anschaulichen] […] can apparently not be drawn so strictly […]’. »

P. Bernays, *Abhandlungen zur Philosophie der Mathematik,* 1976.

Cavaillès on the Mathematical Expansion of Intuition

« The conditions imposed by the strict observance finitists on mathematical existence show a misunderstanding of what makes the very reality of the object. There is no absolute at the outset: what can be said of the point, the continuum, the integer, except that they are elements and origins of sequences in which they take on their meaning and which exceed them? An irreducible intuition is only a thoughtless halt. The Kantian theory of mathematics still obsesses logicians and mathematicians, but because of its superficiality and its illegitimate interpretation in psychological terms. Taken in its authenticity, it remains inadequate in so far as it confuses the dialectical moment of the position of the concept and the transcendental moment of its schematisation. [...] At each stage of mathematical progress, new concepts are denied their right to exist, because they cannot provide a representation in the pre-existing intuitive system. This was the case for the negative numbers, for the imaginaries, for infinitesimals, and for the infinite at the time of Kant. The solution each time is not, as is superficially believed, the discovery of a translation in an immutable spatial intuition (representation of negative numbers on a straight line, of imaginary numbers in the plane) [...]. The solution consists in a transformation of the intuitive zone, i.e. of the rules which lay down the use of […] the schematic system. [...] In other words, intuition in its quiddity progresses in parallel with the dialectical sequence of concepts. [...] [T]here is no absolute transcendental […]. »

J. Cavaillès, 'Transfini et continu', in *Œuvres Completès de Philosophie des Sciences,* Paris: Hermann, 1994.

From « blind or symbolic » knowledge to « intuitive » knowledge

A syntax can be enriched

by adding new symbols (e.g. ω, i)

whose « intuitive » meaning

— far from relying on an *a priori* intuition —

are progressively constituted *a posteriori*

by a « blind » or purely « symbolic » manipulations

according to the rules they obey.

***Blind* or *symbolic knowledge* vs *intuitive knowledge* in Leibniz**

« […] we don't usually grasp the entire nature of a thing all at once […] but in place of the things themselves we make use of signs […] I usually call such thinking, which is found both in algebra and in arithmetic and, indeed, almost everywhere, *blind* or *symbolic*. And indeed, when a notion is very complex, we cannot consider all of its component notions at the same time. When we can […] I call knowledge *intuitive*. »

G.W. Leibniz, *Meditations on Knowledge, Truths, and Ideas*, in *Philosophical Essays*, pp.24-5.

By adding solutions to the $\mathbb{N}$-equations

$$x + n = 0$$

for all $n \in \mathbb{N}$

(which amounts to introduce an additive inverse

for each $n \in \mathbb{N}$, denoted $-n$)

we define the entire numbers

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}.$$

Algebraic Structure of $\mathbb{Z}$

$(\mathbb{Z}, +, \times, 0, 1)$ is a ***ring***.

$(\mathbb{Z}, +, 0)$ is a group (rather than a monoid) since the addition $+$ has now an inverse.

$(\mathbb{Z}, \times, 1)$ is a monoid, since the multiplication $\times$ does not have an inverse.

Whereas the idea of introducing a number

associated with nothing (the *zero*)

was already a non-trivial conceptual leap,

the idea of introducing ***negative numbers***

representing quantities that are « ***less than nothing*** »

met with strong resistance

well into the modern period.

Conceptual History of Negative Numbers (I)

- Diophantus (200 - c.284 AC, *Arithmetica*) rejected negative solutions.

- Fibonnacci (XII century): interpretation of negative quantities in terms of debts.

- Cardano, *Ars Magna,* 1545 (XVI century). Even if he made a systematic use of negative numbers, he considered them as fictions (*numeri ficti*) and he wrote:

« Subtraction is made only of the smaller from the bigger. In fact, it is entirely impossible to subtract a bigger number from a smaller one. »

Conceptual History of Negative Numbers (II)

- In his *Géometrie* (1637), Descartes called negative solutions of equations "false".

- « […] too much truth is paralyzing (I know some who cannot understand that to take four from nothing leaves nothing). »

B. Pascal, *Pensées*, Section II, fr. 72, 1670.

- Coordinates on a line in one of the two possible directions with respect to an arbitrary zero, like dates before Christ (John Wallis, 1616 - 1703):

« But it is also Impossible, that any Quantity […] can be Negative. Since that is not possible that any Magnitude can be Less than Nothing, or any Number Fewer than None. Yet is not that Supposition (of Negative Quantities) either Unuseful or Absurd; when rightly understood. […] when it comes to a Physical Application, it denotes as Real a Quantity as if the Sign were +; but to be interpreted in a contrary sense. »

J. Wallis, *Algebra*, 1673, in D.E. Smith, A Source Book in Mathematics, Dover, 1959.

Cf. D. Rabouin, *Negatives as fictions in 16th and 17th century mathematics*, Hist. Math. 69 (2024) 41-61.

Enlightening the Negative Numbers

« NÉGATIF, adj. (*Algeb.*) quantités *négatives, en Algèbre*, sont celles qui sont affectées du signe -, & qui sont regardées par plusieurs mathématiciens, comme plus petites que zéro. Cette dernière idée n'est cependant pas juste, comme on le verra dans un moment. [...] Il faut avouer qu'il n'est pas facile de fixer l'idée des quantités négatives, & que quelques habiles gens ont même contribué à l'embrouiller par les notions peu exactes qu'ils en ont données. Dire que la quantité *négative* est au-dessous du rien, c'est avancer une chose qui ne se peut pas concevoir. [...] Quand on considère l'exactitude & la simplicité des opérations algébriques sur les quantités *négatives*, on est bien tenté de croire que l'idée précise que l'on doit attacher aux quantités négatives doit être une idée simple, & n'être point déduite d'une métaphysique alambiquée. Pour tâcher d'en découvrir la vraie notion, on doit d'abord remarquer que les quantités qu'on appelle *négatives*, & qu'on regarde faussement comme au-dessous du zéro, sont très souvent représentées par des quantités réelles, comme dans la Géométrie, où les lignes *négatives* ne different des positives que par leur situation à l'égard de quelque ligne au point commun. »

D'Alembert, *Encyclopedie*, 1er ed. (1765), article *Négatif*, Tome 11, pp. 72-4.

Conceptual History of Negative Numbers (III)

• I. Kant, *Attempt to Introduce the Concept of Negative Magnitudes into Philosophy* (1763):

Interpretation of negative numbers

in terms of « real oppositions »

— e.g. forces pointing in opposite directions —

that produce real effects

— e.g. a state of uniform motion —

in contrast with « logical oppositions »

(where the affirmation of both A and $\neg A$ is just a contradiction).

Kant's remark points to a new use of numbers beyond

counting

ordering

measuring (e.g. lengths)

labeling or **coordinating** (e.g. coordinates on the real line),

namely,

changing the orientation (e.g. of a force vector)

by multiplying by negative number.

This last use of numbers leads to the use of numbers

as **scalars in a vector space**.

The cardinality of $\mathbb{Z}$ is $\aleph_0$

($\mathbb{Z}$ is countable)

Indeed we can establish a bijection between $\mathbb{N}$

and $\mathbb{Z}$ ordered as

$0,1,-1,2,-2,3,-3,...$

Hence, the intuition that there are twice

as many whole numbers $0,1,2,3,...$

as entire numbers $...-2,-1,0,1,2,...$

is misleading

(once again, a strict part has the same « size » as the whole).

Rational Numbers

The $\mathbb{Z}$-equation

$$2x - 1 = 0$$

does not have a solution in $\mathbb{Z}$

since the multiplication $\times$ does not have an inverse.

By adding solutions to any $\mathbb{Z}$-equation of the form

$$nx - 1 = 0$$

— which amounts to introduce multiplicative inverses

for each $n \in \mathbb{Z}$, denoted n^{-1}—

we define the ***rational numbers***.

Representation of Rational Numbers as Fractions

Formally, a ***rational number*** $q \in \mathbb{Q}$

is an equivalence class of pairs

(n, m) with $n, m \in \mathbb{Z}$

— usually denoted $\frac{n}{m}$ —

where the equivalence relation $\sim$ is given by

$(n, m) \sim (n', m')$ if $nm' = n'm$.

Example: $\frac{1}{2}$ and $\frac{3}{6}$ are *fractions* that represent the same rational number

since $1 \times 6 = 3 \times 2$.

Positional Representation of Rational Numbers With Respect to a Given Base

For instance, the rational number represented by the fraction $\frac{2536}{100}$

admits a ***decimal positional representation*** by writing it as

$$2 \times \mathbf{10^{1}} + 5 \times \mathbf{10^{0}} + 3 \times \mathbf{10^{-1}} + 6 \times \mathbf{10^{-2}}$$

We could say that the following sequence of numbers between 0 and 9

$$(\ldots,0,2,5.3,6,0,\ldots)$$

— usually written in decimal notation 25,36 —

are « coordinates » of $\frac{2536}{100}$ in the « base »

$$(\ldots,10^{2},10^{1},10^{0}.10^{-1},10^{-2},10^{-3},\ldots).$$

The decimal representation

of a rational number

is not intrinsic

since it depends on the chosen base-10.

If we change the base to (for instance) a base-2

$$(\ldots,2^3,2^2,2^1,2^0.2^{-1},2^{-2},2^{-3},\ldots)$$

or

$$(\ldots 8,4,2,1.\frac{1}{2},\frac{1}{4},\frac{1}{8},\ldots)$$

the coordinates of the corresponding rational number $q \in \mathbb{Q}$

will change accordingly to a sequence of 0's and 1's.

For instance, the rational number $\frac{925}{100}$ can be rewritten as

$$1 \times \mathbf{2^3} + 0 \times \mathbf{2^2} + 0 \times \mathbf{2^1} + 1 \times \mathbf{2^0} + 0 \times \mathbf{2^{-1}} + 1 \times \mathbf{2^{-2}}$$

or

$$(\ldots,0,0,1,0,0,1.0,1,0,0,...)$$

or

$$1001.01$$

Algebraic Structure of $\mathbb{Q}$

$(\mathbb{Q}, +, \times, 0, 1)$ is a ***field***

since both $+$ and $\times$

have now inverses

with respect to the neutral elements 0 and 1 respectively.

How many rational numbers are?

Or, what is the cardinality of $\mathbb{Q}$?

Cantor's Snake

In spite of its density,

the cardinality of $\mathbb{Q}$ is still $\aleph_0$

i.e. $\mathbb{Q}$ is countable:

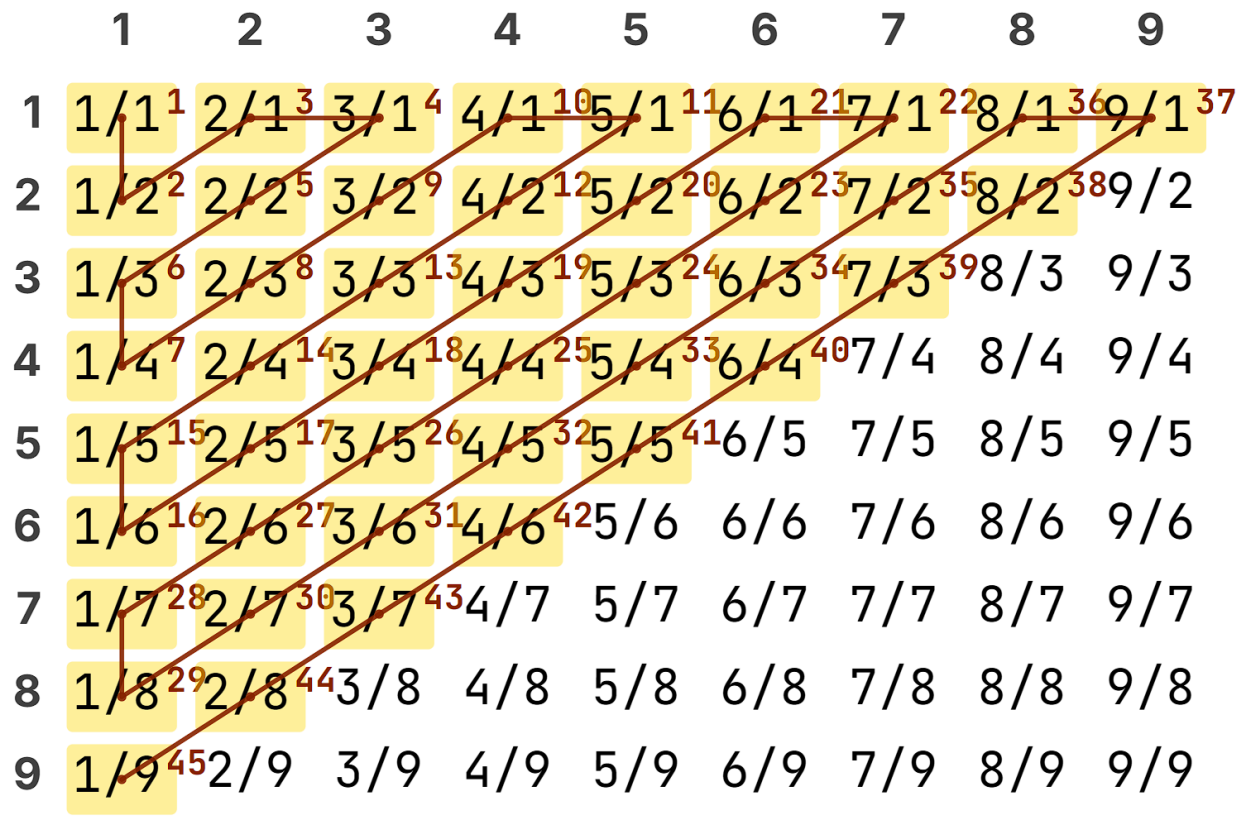
there are as many fractions

$$\pm\frac{n}{m}$$

as natural numbers n.

	1	2	3	4	5	6	7	8	9
1	1/1	2/1	3/1	4/1	5/1	6/1	7/1	8/1	9/1
2	1/2	2/2	3/2	4/2	5/2	6/2	7/2	8/2	9/2
3	1/3	2/3	3/3	4/3	5/3	6/3	7/3	8/3	9/3
4	1/4	2/4	3/4	4/4	5/4	6/4	7/4	8/4	9/4
5	1/5	2/5	3/5	4/5	5/5	6/5	7/5	8/5	9/5
6	1/6	2/6	3/6	4/6	5/6	6/6	7/6	8/6	9/6
7	1/7	2/7	3/7	4/7	5/7	6/7	7/7	8/7	9/7
8	1/8	2/8	3/8	4/8	5/8	6/8	7/8	8/8	9/8
9	1/9	2/9	3/9	4/9	5/9	6/9	7/9	8/9	9/9


1 2 3 4 5 6 7 8 9
1 1/1 1 2/1 3 3/1 4 4/1 10 5/1 6/1 7/1 8/1 9/1
2 1/2 2 2/2 5 3/2 9 4/2 5/2 6/2 7/2 8/2 9/2
3 1/3 6 2/3 8 3/3 4/3 5/3 6/3 7/3 8/3 9/3
4 1/4 7 2/4 3/4 4/4 5/4 6/4 7/4 8/4 9/4
5 1/5 2/5 3/5 4/5 5/5 6/5 7/5 8/5 9/5
6 1/6 2/6 3/6 4/6 5/6 6/6 7/6 8/6 9/6
7 1/7 2/7 3/7 4/7 5/7 6/7 7/7 8/7 9/7
8 1/8 2/8 3/8 4/8 5/8 6/8 7/8 8/8 9/8
9 1/9 2/9 3/9 4/9 5/9 6/9 7/9 8/9 9/9


1 2 3 4 5 6 7 8 9
1 1/1 1 2/1 3 3/1 4 4/1 10 5/1 11 6/1 21 7/1 22 8/1 36 9/1 37
2 1/2 2 2/2 5 3/2 9 4/2 12 5/2 20 6/2 23 7/2 35 8/2 38 9/2
3 1/3 6 2/3 8 3/3 13 4/3 19 5/3 24 6/3 34 7/3 39 8/3 9/3
4 1/4 7 2/4 14 3/4 18 4/4 25 5/4 33 6/4 40 7/4 8/4 9/4
5 1/5 15 2/5 17 3/5 26 4/5 32 5/5 41 6/5 7/5 8/5 9/5
6 1/6 16 2/6 27 3/6 31 4/6 42 5/6 6/6 7/6 8/6 9/6
7 1/7 28 2/7 30 3/7 43 4/7 5/7 6/7 7/7 8/7 9/7
8 1/8 29 2/8 44 3/8 4/8 5/8 6/8 7/8 8/8 9/8
9 1/9 45 2/9 3/9 4/9 5/9 6/9 7/9 8/9 9/9

Since n can be as large as we wish, its inverse

$$\frac{1}{n}$$

can be as small as we wish.

Thus, we have introduced numbers

that can be *potentially infinitely small*

just as n can be *potentially infinitely large*.

However, because there is no largest n,

there can be no smallest $\frac{1}{n}$.

$\mathbb{Q}$ is dense

Between any two rational numbers,

there is an infinite number

of other rational numbers.

Demonstration.

Given two rational numbers $q < p \in \mathbb{Q}$,

it is always possible to choose n big enough

$$\text{so that } \frac{1}{n} < p - q.$$

$$\text{Hence, } q < q + \frac{1}{n} < p.$$

The ***continuous***

is that which is not ***discrete***,

like a line traced without ever lifting the pen from the paper,

a set in which no gap separates any two elements,

a gapless extension.

« […] ese laberinto, que consta de una sola línea recta y que es indivisible, incesante. »

J.L. Borges, *La muerte y la brújula*.

By adding numbers between any two numbers

and iterating this operation,

it seems that we can fill the line.

Can mere *density*

endow a set of points with

the nature of a ***continuum***

that is, of a ***gapless extension***?

« When points are situated in such a way that there are no two between which there is not an intermediate point, then, by that fact, there is continuous extension. »

G.W. Leibniz, Letter to Des Bosses, May 29, 1716.

« If we try to form a clear idea of what we call a '*continuous extension*' or '*continuum*', we are forced to declare that a continuum is present when, and only when, we have an aggregate of simple entities (instants or points or substances) so arranged that each individual member of the aggregate has, at each individual and sufficiently small distance from itself, at least one other member of the aggregate for a neighbour. »

B. Bolzano, *Paradoxes of the Infinite*, 1851, p.129.

However… Rational Density is Not Enough to Guarantee Continuity

« Of the greatest importance, however, is the fact that in the straight line L there are infinitely many points which correspond to no rational number. […] the ancient Greeks already knew and had demonstrated that there are lengths incommensurable with a given unit of length, e. g., the diagonal of the square whose side is the unit of length. If we layoff such a length from the point o upon the line we obtain an end-point which corresponds to no rational number. […] The above comparison of the domain $\mathbb{Q}$ of rational numbers with a straight line has led to the recognition of the existence of gaps, of a certain incompleteness or discontinuity of the former, while we ascribe to the straight line completeness, absence of gaps, or continuity. In what then does this continuity consist? »

R. Dedekind, *Continuity and Irrational Numbers*, 1872, pp.8-10.

Irrational numbers

We can rephrase the Greeks' discovery of irrational numbers

by saying that there are $\mathbb{Q}$-polynomial equations

— e.g. $x^2 - 2 = 0$ —

whose solutions are not in $\mathbb{Q}$,

i.e. cannot be written as $\frac{n}{m}$

for integers $n, m \in \mathbb{Z}$.

There are *algebraic numbers*

— numbers that are solutions of $\mathbb{Q}$-equations —

that are not rational.

Continuous Fractions

Even if « irrational » numbers cannot be expressed as fractions, they can be nonetheless be expressed in a very « rational » form as « continuous fractions », like for instance:

$$\sqrt{2} = 1 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \dots}}}}$$

$$\varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \dots}}}}} \quad \text{(golden ratio)}$$

Hence, ***density***

— the fact that between two rational numbers

there are infinite rational numbers —

does not guarantee ***completeness*** (or *gaplessness*).

ℚ is full of (algebraic) gaps.

Transcendental Gaps in $\mathbb{Q}$

Besides $\mathbb{Q}$-polynomial equations with no $\mathbb{Q}$-solutions (algebraic gaps)

there other kinds of « symptoms »

of the limitations of $\mathbb{Q}$,

like sequences $a_0, a_1, a_2, \ldots$ of rational numbers

that verify ***Cauchy's condition***

— that for any $\epsilon > 0$, there is a positive integer N

such that for all $m, n > N$ we have

$$|a_m - a_n| < \epsilon,$$

which means that the difference

between terms tends to 0 as n increases —

but that do not converge to any algebraic number.

Gaps in $\mathbb{Q}$ that are not algebraic are called *transcendental*.

Zeno's First Paradox: The Dichotomy

« The first [paradox] asserts the non-existence of motion on the ground that that which is in locomotion must arrive at the half-way stage before it arrives at the goal. »

Aristotle, *Physics*, 239b11-13.

« [...] if there is motion, it is necessary that a moving thing will entirely traverse an infinite number of things in a finite time; but this is impossible; consequently, there is no motion. »

Simplicius, *Commentary on Aristotle's* Physics, 1013,3-14.

Zeno's Arrow

The $\mathbb{Q}$-sequence

$$a_n = \sum_{k=1}^{n} \frac{1}{2^k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \cdots$$

where $a_1 = 0.5, a_2 = 0.75, a_3 = 0.875, a_3 = 0.9375, \ldots$

converges to the rational number 1.

Generalized Zeno's Arrows with Ghostly Targets

There are convergent sequences a_n of rational numbers

— which we might call ***generalized Zeno's arrows*** —

that do not converge to any algebraic number.

These generalized Zeno's arrows strike $\mathbb{Q}$-nonexistent targets.

Examples of Generalized Zeno's Arrows striking Ghostly Targets

The $\mathbb{Q}$-sequence

$$a_n = \sum_{k=0}^{n} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots$$

tends to a « transcendental » (i.e. non-algebraic) number denoted e.

The $\mathbb{Q}$-sequence

$$a_n = 4 \sum_{k=0}^{n} \frac{(-1)^k}{2k+1} = 4(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots)$$

tends to a « transcendental » (i.e. non-algebraic) number denoted π.

Construction of the Real Numbers $\mathbb{R}$ from Cauchy Sequences

Every sequence a_n of rational numbers that verifies ***Cauchy's condition***

— that for any $\epsilon > 0$, there is a positive integer N

such that for all $m, n > N$ we have

$$|a_m - a_n| < \epsilon$$ —

defines a ***real number***.

Every generalized Zeno's arrow becomes the index of a « real » target.

The decimal expansion of any real number x

defines a Cauchy sequence (x_n)

whose n-th term x_n is the truncation to n

of the decimal expansion of x.

Example: π is given by the Cauchy sequence 3,3.1,3.14,3.141,3.1415,...

Dedekind's argument against the completeness of $\mathbb{Q}$.

There are partitions of the set $\mathbb{Q}$ of rational numbers

into two nonempty classes

L (the *left set* of the cut)

— without a largest element —

and R (the *right set* of the cut)

— without a smallest element —

such that every member L is smaller than every member R

$l < r$ for every $l \in L$ and $r \in R$

and yet no rational number lies between the two classes.

Such a « Dedekind cut » is a manifestation of a gap in $\mathbb{Q}$, i.e. of the fact that $\mathbb{Q}$ is not a gapless continuum.

Such a « Dedekind cut » associated to a gap in $\mathbb{Q}$ defines an irrational number.

$\sqrt{2}$ as a Dedekind cut

For instance, $\mathbb{Q}$ can be « cut » in the two sets

$$L = \{q / q < 0 \text{ or } q^2 < 2\}$$

and

$$R = \{q / q > 0 \text{ and } q^2 > 2\},$$

but there is no rational number that is

larger than every member of L

and smaller than every member of R.

Such a « cut » defines a gap in $\mathbb{Q}$.

Construction of the Real Numbers $\mathbb{R}$ from Dedekind's Cuts

The set of real numbers $\mathbb{R}$

can be defined as the set of ***Dedekind cuts*** of $\mathbb{Q}$,

where a Dedekind cut of $\mathbb{Q}$

is a non-empty proper subset L of $\mathbb{Q}$ such that

- **Downwards closeness**: for all $q \in L$, if $p \in \mathbb{Q}$ and $p < q$, then $p \in L$.
- **No maximal element**: if $q \in L$, there exists $q' \in L$ such that $q < q'$.

Every rational number q is defined by the Dedekind cut

$$L_q = \{p \in \mathbb{Q} / p < q\}$$

Real Numbers

are either

algebraic

—numbers that are solutions to polynomial equations —

like $\sqrt{2}$ and the golden ratio φ, etc.;

or ***transcendental***

— real numbers that are not solutions to polynomial equations —

like π or e.

In turn, there are real numbers

that are *computable* (e.g. $\sqrt{2}, \pi, e$)

— i.e. numbers that can be approximated

to any desired precision

by a finite, effective algorithm —

and others that are *non-computable*.

Having departed from the shores

of the « natural » and the « rational »

to embrace the « fictitious » and the « irrational »,

we now find ourselves in the realm of the « real »

that is, the gapless,

that which unfolds seamlessly,

without ever needing to leap over a discontinuity.

The Continuum Hypothesis

How many points are there

on the real line $\mathbb{R}$?

Moreover, Cantor proved in 1874

that the algebraic numbers

— including irrationals like $\sqrt{2}$ —

are also countable.

Idea of the demonstration:

- There are $\mathbb{Z}^{n+1}$polynomials with coefficients in $\mathbb{Z}$ of degree $n \geqslant 1$, each of which has at most n solutions.

- This yields a countable number $n\mathbb{Z}^{n+1}$ of algebraic numbers arising as solutions of polynomial equations of degree n (since a Cartesian product of countable sets is countable).

- The union of all these countable numbers for all n is countable.

Thus, even when we leave the shores of $\mathbb{Q}$

to embrace all the algebraic numbers,

we are still within the realms of the countable.

But there are still the transcendental numbers

to might spoil the countability of $\mathbb{R}$…

G. Cantor proved (1874, 1891) that

The *continuum*

is strictly greater

than the *countable/discrete*.

In other terms, it is not possible to establish

a bijection between $\mathbb{R}$ and $\mathbb{N}$.

First Proof of the Uncountability of $\mathbb{R}$

(G. Cantor, *Über eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen*, 1874)

- **Cantor's Excess Theorem**: the cardinal $|\mathscr{P}(X)|$ of the set of parts (subsets)

$$\mathscr{P}(X) = \{0,1\}^X$$

of a set X

(where $\{0,1\}^X$ is the set of maps $X \to \{0,1\}$)

is bigger than the cardinal $|X|$ of X:

$$|\mathscr{P}(X)| = 2^{|X|} > |X|$$

that is, there are always more parts (subsets) than elements.

- **« The continuum […] is the set of parts of the discrete »** (A. Badiou, *Being and Event*):

$$\mathbb{R} = \mathscr{P}(\mathbb{N}) = \{0,1\}^{\mathbb{N}}$$

- Hence, the continuum is greater than the countable:

$$|\mathbb{R}| = 2^{\aleph_0} > |\mathbb{N}| = \aleph_0$$

Excess of Parts over Elements for Finite Sets

For finite sets

$$|\mathscr{P}(X)| = 2^{|X|}$$

i.e., the number of parts $|\mathscr{P}(X)|$ grows exponentially

with the number of elements $|X|$.

A part A of X can be defined by specifying

if each $x \in X$ belongs or not to A,

which defines a « characteristic function » $\chi_A : X \to \{0,1\}$

such that $\chi_A(x) = 1$ if $x \in A$ and $\chi_A(x) = 0$ if $x \notin A$.

Now, there are $2 \times \ldots \times 2$ (times the number of elements of X) $= 2^{|X|}$ functions $X \to \{0,1\}$.

Since $2^{|X|} > |X|$, the number of parts $|\mathscr{P}(X)|$ is always greater

than the number of elements $|X|$ for X finite.

Cantor's Excess Theorem

or Why there are always more parts than elements

- Let's assume a one-to-one correspondence

$$S : X \to \mathscr{P}(X)$$

between elements and parts.

- Let's define the following part of X:

$$Q = \{x \in X / x \notin S(x)\}$$

- Since S is assumed to be one-to-one, there should be a $q \in X$ such that $S(q) = Q$.

- Does q belong to Q?

  .q cannot be in $Q = S(q)$ since Q contains only the elements x that are not in $S(x)$.

  .But if q is not in Q, then it should be in Q since Q contains all the elements x that are not in $S(x)$.

- Hence, a one-to-one correspondence between X and $\mathscr{P}(X)$ cannot exist.

- $X \subset \mathscr{P}(X)$ where the inclusion if given by $x \mapsto \{x\}$.

Consequence of Cantor's Excess Theorem

Since the set $\mathscr{P}(X)$ of parts/subsets of X

has more elements than X

— even when X is infinite —,

there is no largest infinity.

The Continuum is the Set of Parts of the Discrete (I)

$$\mathbb{R} = \mathscr{P}(\mathbb{N}) = \{0,1\}^{\mathbb{N}}$$

which implies:

$$|\mathbb{R}| = 2^{|\mathbb{N}|} = 2^{\aleph_0}$$

This can be proved by showing that

$$\mathbb{R} \hookrightarrow \mathscr{P}(\mathbb{N})$$

and that

$$\mathscr{P}(\mathbb{N}) \hookrightarrow \mathbb{R},$$

which implies

— by the by the Cantor–Schröder–Bernstein theorem —

$$\mathbb{R} = \mathscr{P}(\mathbb{N}).$$

The Continuum is the Set of Parts of the Discrete (II)

Proof that $\mathbb{R} \hookrightarrow \mathscr{P}(\mathbb{N})$

Since $\mathbb{Q}$ is countable,

there is a bijection

$$b : \mathbb{Q} \to \mathbb{N}.$$

Each $x \in \mathbb{R}$ defines the following subset of $\mathbb{N}$:

$$A_x = b(\{q \in \mathbb{Q} / q < x\})$$

In other terms, we take the Dedekind cut that represents x (which is a set of rationals)

and then use the bijection

$$b : \mathbb{Q} \to \mathbb{N}$$

to convert this set of rationals into a set of natural numbers, i.e. a part of $\mathbb{N}$.

The Continuum is the Set of Parts of the Discrete (II)

Proof that $\mathscr{P}(\mathbb{N}) \hookrightarrow \mathbb{R}$

Each subset

$$A \subset \mathbb{N}$$

defines a « characteristic function

$$\chi_A : \mathbb{N} \to \{0,1\},$$

i.e. an infinite sequence of 0's and 1's indexed by $\mathbb{N}$.

Each such sequence defines a real number

in the interval [0,1] written in base-3 as

$$0.\chi_A(0)\chi_A(1)\chi_A(2)\chi_A(3)\ldots$$

Second Proof of the Uncountability of $\mathbb{R}$

Cantor's Diagonal Argument

(*Über eine elementare Frage der Mannigfaltigkeitslehre,* 1891)

- Let's assume that it is possible to establish a one-to-one correspondence between the real numbers in the interval [0,1] and $\mathbb{N}$.

- This means that it is possible to make a list indexed by $\mathbb{N}$ of all $x \in [0,1]$.

- It is always possible to construct « diagonal » real numbers in the interval [0,1] that are not in the list.

List all real numbers between 0 and 1

1) 0.81728056931661013616090881330367580918331390128519747920…
2) 0.58787619690593538894245451824849205989889286583490896030…
3) 0.33327924801963223542552639721823659830585587541248796990…
4) 0.65513924153272811245900919954349430689648453470508978210…
5) 0.66560141186785141558028741544170954400044602585952229340…
6) 0.07772339164214430688263308157460042035244021202327992180…
7) 0.03821153525967325839348345297560221305429837867916572150…
8) 0.21951900414452078193905288558439117290841235286267143230…
9) 0.90602421167967824425409891543774832733240440346204144550…
10) 0.71788275612658991468952573508616920659669778308438007900…
11) 0.13184577057263264368716835976498388364636780620729660200…
12) 0.88105502948384574948184695942626437579151752259136938120…
13) 0.81019501082193672630738755035503061529558794514296463800…
14) 0.12876529962860790788912142446412832571767103179130564000…
15) 0.78070719203767892711304362606160498892216148913396369520…
16) 0.84994003422777524274714141283138509286627529374312003950…
17) 0.26200197034813093277282836237523147304415860495136885220…
18) 0.71527255403422136941506293794252051103341692070235909400…
19) 0.04867539094699200058213078457796579497911324824851009810…
20) 0.40493497450073615836984915388854910799132956274007248840…
… …

Select the digits on the diagonal

1) 0.**8**1728056931661013616090881330367580918331390128519747 92…
2) 0.5**8**7876196905935388942454518248492059898892865834908960 3…
3) 0.33**3**279248019632235425526397218236598305855875412487969 9…
4) 0.655**1**392415327281124590091995434943068964845347050897821…
5) 0.6656**0**1411867851415580287415441709544000446025859522293 4…
6) 0.07772**3**391642144306882633081574600420352440212023279921 8…
7) 0.038211**5**3525967325839348345297560221305429837867916572 15…
8) 0.219519**0**0414452078193905288558439117290841235286267143 23…
9) 0.906024211**6**7967824425409891543774832733240440346204144 55…
10) 0.71788275**6**1265899146895257350861692065966977830843800 790…
11) 0.131845770**5**726326436871683597649838836463678062072966 020…
12) 0.8810550294**8**38457494818469594262643757915175225913693 812…
13) 0.810195010821**9**3672630738755035503061529558794514296463 80…
14) 0.12876529962860**7**907889121424464128325717671031791305640 0…
15) 0.780707192037678**9**27113043626061604988922161489133963695 2…
16) 0.84994003422777**5**2427471414128313850928662752937431200395…
17) 0.26200197034813093**2**77282836237523147304415860495136885 22…
18) 0.715272554034221369**4**150629379425205110334169207023590940…
19) 0.048675390946992000**5**821307845779657949791132482485100981…
20) 0.40493497450073615836**9**8491538885491079913295627400724884…
… …

Add 1 to the digits : 0.99421461228401934067 —> Not in the list

1) 0.917280569316610136160908813303675809183313901285197479 2…
2) 0.597876196905935388942454518248492059898892865834908960 3…
3) 0.334279248019632354255263972182365983058558754124879699…
4) 0.655239241532728112459009199543494306896484534705089782 1…
5) 0.665611411867851415580287415441709544000446025859522293 4…
6) 0.077724391642144306882633081574600420352440212023279921 8…
7) 0.038211635259673258393483452975602213054298378679165721 5…
8) 0.219519014144520781939052885584391172908412352862671432 3…
9) 0.906024212679678244254098915437748327332404403462041445 5…
10) 0.717882756226589914689525735086169206596697783084380079 0…
11) 0.131845770582632643687168359764983883646367806207296602 0…
12) 0.881055029484845749481846959426264375791517522591369381 2…
13) 0.810195010821036726307387550355030615295587945142964638 0…
14) 0.128765299628617907889121424464128325717671031791305640 0…
15) 0.780707192037679927113043626061604988922161489133963695 2…
16) 0.849940034227775342747141412831385092866275293743120039 5…
17) 0.262001970348130942772828362375231473044158604951368852 2…
18) 0.715272554034221360415062937942520511033416920702359094 0…
19) 0.048675390946992000682130784577965794979113248248510098 1…
20) 0.404934974500736158379849153888549107991329562740072488 4…
… …

The real number thus constructed

(by adding 1 to the elements in the diagonal)

is not in the original list since

- it differs from the first digit of the first number,
- it differs from the second digit of the second number,
- it differs from the n digit of the nth number.

The continuum $\mathbb{R}$ is bigger that the countable/discrete,

i.e. there are more real numbers than natural numbers.

Now, what is the cardinality $|\mathbb{R}|$ of the *continuum* $\mathbb{R}$?

How bigger is it?

$\aleph_1$

• Let's consider the first ordinal — denoted ω_1 — that is not countable (i.e. isomorphic to $\mathbb{N}$).

• Since every ordinal can be understood as the collection of all the preceding ones, ω_1 is the collection of all countable ordinals.

• $\aleph_1$ is the cardinal of ω_1.

The **Continuum Hypothesis** (Cantor, 1878) states that

$$|\mathbb{R}| = \aleph_1$$

i.e. that the cardinality of the continuum $|\mathbb{R}|$

is the smallest cardinal after $\aleph_0$.

The validity of this conjecture

is independent of ZFC-set theory.

(K. Gödel, 1940; P. Cohen, 1963).

We do not know how many real numbers are.

Philosophical Interpretations of the Continuum

Main antagonism in the interpretation of the continuum

- Either the continuum as such is essentially inintelligible and has to be constructed out of discrete entities (Cantor) or postulated as a kind of pre-mathematical a priory (Brouwer). Only the discrete is intelligible.

- Or the continuum is the very condition of intelligibility (Leibniz), making for instance possible the predictive capacities (via differential and integral calculus) of modern science. Only the continuous is intelligible.

I. **The continuum is made of points**

The continuum is made of

discrete extensionless point-like « elements »,

i.e. a continuous extension can be generated by an accumulation of points

(Bolzano, Cauchy, Dedekind, Cantor).

The arithmetization of love according to Proust

“Car ce que nous croyons notre amour, notre jalousie, n’est pas une même passion continue, indivisible. Ils se composent d’une infinité d’amours successifs, de jalousies différentes et qui sont éphémères, mais par leur multitude ininterrompue donnent l’impression de la continuité, l’illusion de l’unité. La vie de l’amour de Swann, la fidélité de sa jalousie, étaient faites de la mort, de l’infidélité, d’innombrables désirs, d’innombrables doutes, qui avaient tous Odette pour objet. S’il était resté longtemps sans la voir, ceux qui mouraient n’auraient pas été remplacés par d’autres. Mais la présence d’Odette continuait d’ensemencer le coeur de Swann de tendresses et de soupçons alternés.”

M. Proust, *Du côté de chez Swann, A la recherche du temps perdu.*

II. The continuum is made of infinitesimals

The continuum is made of « elements »

which are not extensionless points,

by rather ***infinitesimals***,

i.e. hypothetical new kind of numbers

that are bigger than zero

but smaller than any real number

(Peirce, Robinson).

III. The continuum is an indivisible unity dynamically generated by motion

« […] a continuum cannot be composed of indivisibles, e.g. a line of points. […] nothing continuous was divisible into partless parts. […] Further, every continuum is divisible into divisibles *ad infinitum*. »

Aristotle, *Physics*, Book VI, 231.

« Diogenes the Cynic, having heard these puzzles once, said nothing in reply to them, but stood up and walked and by means of self-evidence itself resolved the paradoxes in the arguments. »

Simplicius, *Commentary on Aristotle's* Physics, 1012,22-1013,2

Newton on Continuous Magnitudes

Continuity is dynamically generated by « fluent quantities »

« I don’t here consider Mathematical Quantities as composed of Parts extremely small, but as generated by a continual motion. Lines are described, and by describing are generated, not by any apposition of Parts, but by a continual motion of Points. […] Time by a continual flux, and so in the rest. These Geneses are founded upon Nature, and are every Day seen in the motion of Bodies. »

I. Newton, *Quadrature of Curves*.

IV. The continuum is beyond symbolic reason

The continuum is a primitive notion

that cannot be fully apprehended by

— a necessarily discrete —

symbolic formalism

(Brouwer, Bergson, Thom).

Brouwer's « Intuitionism » (1881-1966)

The continuum is an *Ur*-intuition

associated with the (the flow of) time

— to the exclusion of space

(as a consequence of the emergence of non-Euclidean geometries) —

as the *a priori* form of experience that generates both

the sequence of natural numbers,

— associated with the « succession » of one sensation after another

(that Brouwer calls « two-oneness ») —

and the *continuum* understood as the in-between of two successive instants.

« However weak the position of intuitionism seemed to be after this period of mathematical development, it has recovered by abandoning Kant's apriority of space but adhering the more resolutely to the apriority of time. This neo-intuitionism considers the falling apart of moments of life into qualitatively different parts, to be reunited only while remaining separated by time, as the fundamental phenomenon of the human intellect, passing by abstracting from its emotional content into the fundamental phenomenon of mathematical thinking, the intuition of the bare two-oneness. This intuition of two-oneness, the basal intuition of mathematics, creates not only the numbers one and two, but also all finite ordinal numbers, inasmuch as one of the elements of the two-oneness may be thought of as a new two-oneness, which process may be repeated indefinitely; this gives rise still further to the smallest infinite ordinal number ω. Finally this basal intuition of mathematics, in which the connected and the separate, the continuous and the discrete are united, gives rise immediately to the intuition of the linear continuum, i.e., of the 'between,' which is not exhaustible by the interposition of new units and which therefore can never be thought of as a mere collection of units. »

L.E.J. Brouwer, *Intuitionism and formalism*, 1912.

Thom on the Continuum (I)

« I would accept that, in any kind of description, we have a discrete element, because a true continuum has no points. We are unable to specify anything in the continuum. The continuum is something which cannot be described. It is a sort of unsayable. It is a world in which one lives outside of symbolic description. But nevertheless, it exists, despite the fact that we cannot describe it in any sense. »

R. Thom, Round-Table discussion, in Casacuberta and Castellet (eds), *Mathematical Research Today and Tomorrow Viewpoints of Seven Fields Medalists*, p.102.

Thom on the Continuum (II)

« People who say that the discrete is more important than the continuous always forget that discrete entities are carved out from within a continuous medium. It is in this sense that there is an ontological primacy of the continuous. [...] People who believe in pure discreteness are obsessed with writing, with the discrete event symbolized by a letter… »

R. Thom, *Théorie des catastrophes, Mathématiques et sciences humaines*, tome 59, 1977, 3-81, p.38.

Thom on the Continuum (III)

« Now, among the primary intuitions, there is one that is essential: that of time. And time, in the flow of its passing, appears to us as fundamentally continuous: between two instants *a* and *b*, one can always imagine an intermediate instant *c*. Consequently, we are led to consider as 'ontologically better founded' a theory such as Differential Calculus, which aims at representing continuous geometric entities—such as curves and surfaces—than a purely discrete theory, like Set-theoretic Logic, for example. This priority of the continuum that we postulate goes against the traditional approach, which treats the real line as a constructed object—by a procedure that is in fact not very constructive—out of the integers and then the rationals (via Dedekind cuts or the completion of Cauchy sequences).

[...] Of the continuum in its pure state, nothing can be said; it is an object as unsayable as divinity may be for a mystic. In order to make the continuum an object of representation, in order to be able to say something about it, one must necessarily make 'marks' in it, trace 'figures' within it, which stand out against an undifferentiated background. Every communicable representation thus involves a discrete figuration. The permanent temptation, then, is to forget the background and keep only the figure: an excessive simplification strongly suggested to us by the discrete character of the phonological combinatorics of human language, and into which logicians—and, following them, quite a few formalist mathematicians—have rushed headlong. »

R. Thom, *Les réels et le calcul différentiel ou la mathématique essentiel, Séminaire de Philosophie et Mathématiques*, 1982, fascicule 3, Les réels et le calcul différentiel, p. 1-20.

Le différentiel en deçà de la representation ou la mystification du calcul

« […] les interprétations finitistes modernes ne trahissent pas moins la nature du différentiel que les anciennes interprétations infinitistes, parce qu'elles laissent échapper toutes deux la source extra-propositionnelle ou sub-représentative, c'est-à-dire le 'problème', d'où le calcul tire son pouvoir. »

G. Deleuze, *Différence et répétition*, p. 339.

Cantor on the (Arithmetization of the) Continuum

« Leucippus, Democritus and Aristotle regard the continuum as a composite, which consists of endlessly divisible particles (*ex partibus sine fine divisibilibus*), whereas Epicurus and Lucretius consider that it is a composite of atoms that are finite things. From this a great dispute later arose among philosophers, some of whom followed Aristotle, while others followed Epicurus, while others, in an attempt to steer clear of the dispute, sided with Thomas Aquinas in the claim that the continuum consists neither of an infinite number nor of a finite number of parts, but of no parts at all. […] Here we observe the *medieval-scholastic* origin of a view that we still find to be held today, according to which the continuum is an indivisible concept or, as others put it, a notion that is purely a priori, and hardly amenable as a basis for any definition. Every attempt at an arithmetical *determination* of this *mystery* is regarded as an inadmissible interference and rejected with all due intensity, and those of a diffident disposition get the impression that the 'continuum' is not a *mathematical-logical notion* but rather a *religious dogma*. […] in my opinion, the use of the *concept of time* or the *intuition of time* is not applicable when discussing the much more fundamental and general concept of the continuum. In my opinion, *time* is a notion which requires for any clear explanation of it, a presupposition of the concept of continuity, a concept that is independent of it. […] It is also my conviction that one cannot do anything with the so-called *form of intuition* of space in order to gain information about the continuum […] Hence I have no choice but to attempt, with the help of the real number concepts defined in § 9, to form a purely arithmetical concept of a point continuum that is as general as possible. »

G. Cantor, *Foundations of a General Theory of Manifolds: A Mathematico-Philosophical*

Continuity as a Rational Principle

Natura non facit saltus

« Nothing takes place suddenly, and it is one of my great and best confirmed maxims that nature never makes leaps. I called this the Law of Continuity […] It implies that any change from small to large, or vice versa, passes through something which is […] in between […] just as one could never traverse a certain line or distance without first traversing a shorter one. […] To think otherwise is to be ignorant of the immeasurable fineness of things, which always and everywhere involves an actual infinity. »

Gottfried Leibniz, *New Essays on Human Understanding*, Preface, 56-57.

Badiou on Leibniz's Continuity Principle

« […] presentation is continuous, and it does not tolerate interruption […]: 'The present is always pregnant with the future and no given state is naturally explainable save by means of that which immediately preceded it. If one denies this, the world would have hiatuses, which would overturn the great principle of sufficient reason, and which would oblige us to have recourse to miracles or to pure chance in the explanation of phenomena.' The event is thus excluded on the following basis: the complete language is the integral calculus […], whilst a local approximation already authorizes its differential calculus. »

A. Badiou, *Being and Event*, Meditation 30: Leibniz, p.319.

Tropical Discontinuities

« You wait for explanations, and, like Descartes, you receive only ever-new pieces of information, without connection, and from sentence to sentence, from word to word, you never know what to expect, ready to throw in the towel, to withdraw from this incessant abortion of the continuous, from this permanent frustration of your expectation […] »

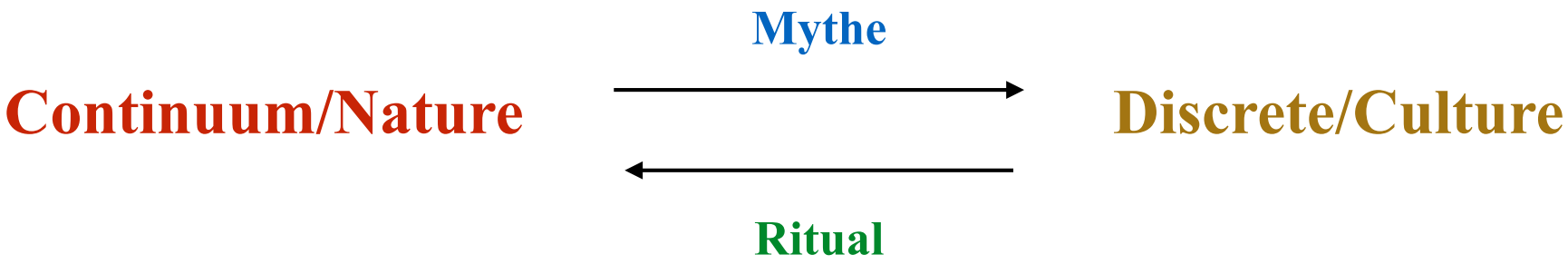
J.-.C. Goddard, *Scabby Black Brazilian,* Urbanomic, 2024.

Declensions of the *Discrete-Continuum* dialectic

- Pitch vs. Rythm (Stokhausen).
- Nature vs. Culture (Lévi-Strauss).
- Singing vs. spoken language.
- Gradualism vs. saltationism in evolutionary biology.
- Bergson vs. Bachelard (*durée* vs. cinematographic model in H. Bergson.
- Smooth Schrödinger evolution vs. quantum leaps (Quantum Mechanics).
- Waves vs. particles (Quantum Mechanics).
- Fields vs. Particles (Quantum Field Theory).

The *continuum/discrete* opposition

in Lévi-Strauss' structural anthropology



Refs.: G. Schrempp, *Magical Arrows: The Maori, the Greeks, and the Folklore of the Universe*, 1992, University of Wisconsin Press; P.A. Lolli, *O contínuo e o discreto em Lévi-Strauss: transformações ameríndias*, Tellus, ano 12, n. 22, p. 81-105, jan./jun. 2012 Campo Grande, MS.

On the *continuum/nature* - *discrete*/*culture* opposition

« I shall establish the existence of an isomorphism between two oppositions; that of nature and culture and that of continuous and discrete quantities. »

C. Lévi-Strauss, *The Raw and the Cooked*, p.28.

« […] the transition from nature to culture corresponds, in native thought, to the transition from the continuous to the discontinuous. »

C. Lévi-Strauss, *The Raw and the Cooked*, p.279.

« It is as if South American thought […] invested chromaticism [« in the very general sense of the use of small intervals »] with a primordial maleficence of such a kind that long intervals, which are necessary in culture for it to exist and in nature for it to be 'thinkable' to men, can only be the result of the self-destruction of a primeval continuity, whose power is still to be felt at those rare points where it has survived its own self-destructiveness […] »

C. Lévi-Strauss, *The Raw and the Cooked*, p.280.

The Mythological Transition from Continuous to Discrete Quantity (I)

« It would therefore seem that the two myths, taken together, refer to three domains, each of which was originally continuous, but into which discontinuity had to be introduced in order that each might be conceptualized. In each case, discontinuity is achieved by the radical elimination of certain fractions of the continuum. Once the latter has been reduced, a smaller number of elements are free to spread out in the same space, while the distance between them is now sufficient to prevent them overlapping or merging into one another. Men had to become less numerous for the most closely related physical types to be clearly distinguishable. For if we admit the existence of clans or tribes bearing *insignificant* gifts—that is whose distinctive originality is as weak as one cares to imagine—we should run the risk of finding that between any two given clans or communities there was an unlimited number of other clans or other tribes, each one of which would differ so slightly from its immediate neighbors that they all would ultimately merge into each other; whereas in any field a system of significances can be constructed only on the basis of discrete quantities. […] In all these instances, therefore, a discrete system is produced by the destruction of certain elements or their removal from the original whole. […] it is clear that the myths I am comparing all offer an original solution to the problem of the change-over from continuous quantity to discrete quantity. »

C. Lévi-Strauss, *The Raw and the Cooked*, pp.52-53.

The Mythological Transition from Continuous to Discrete Quantity (II)

« […] mythic thought proceeds in the same way as language, which chooses phonemes from among the natural sounds of which a practically unlimited range is to be found in childish babbling. For, as in the case of language, the empirical material is too abundant to be all accepted indiscriminately or to be all used on the same level. […] it must be whittled down. Only a few of its elements are retained—those suitable for the expression of contrasts or forming pairs of opposites. But, as with language, the discarded elements are not thereby eliminated. They are always present in latent fashion, behind those that have been singled out […] the virtually unlimited totality of the elements always remains available. »

C. Lévi-Strauss, *The Raw and the Cooked*, p.341.

The Ritualistic Transition from Discrete to Continuous Quantity (I)

« Pour acquérir un rituel […] il a fallu qu'un humain abjurât les distinctions claires et nettes en vigueur dans la culture et la société ; que […] il revînt à l'état de nature marqué par la promiscuité des sexes et la confusion entre les degrés de parenté […] Ainsi donc, tandis que le mythe tourne résolument le dos au continu pour découper et désarticuler le monde au moyen de distinctions, de contrastes et d'oppositions, le rite suit un mouvement en sens inverse : parti des unités discrètes qui lui sont imposées par cette conceptualisation préalable du réel, il court après le continu et cherche à le rejoindre, bien que la rupture initiale opérée par la pensée mythique rende la tâche impossible à jamais. […] le rite se réfère bien à ces catégories, lois ou axiomes. Mais le rite ne les crée pas et s'emploie plutôt, sinon à les désavouer, au moins à oblitérer temporairement les distinctions et oppositions qu'ils édictent, en faisant apparaître entre celles-ci toutes sortes d'ambiguïtés, de compromis et de passages. »

C. Lévi-Strauss, *Mythologiques 4 : L'homme nu.*

The Ritualistic Transition from Discrete to Continuous Quantity (II)

« […] pour ce qui est du rituel, l'anxiété accompagnatrice […] tient […] à la crainte que les découpages opérés sur le réel par la pensée discrète en vue de le conceptualiser ne permettent plus de rejoindre […] la continuité du vécu. Donc une anxiété qui […] résulte de ce que le pensé, du seul fait qu'il est du pensé, creuse un écart croissant entre l'intellect et la vie. Le rituel n'est pas une réaction à la vie, il est une réaction à ce que la pensée a fait d'elle. »

C. Lévi-Strauss, *Mythologiques 4 : L'homme nu.*

Complex Numbers

History of Complex Numbers

- Gerolamo Cardano, *Ars Magna* (1545).

- Raphael Bombelli, *L'Algebra* (1572).

- Descartes, *La géométrie* (1637): introduction of the term « imaginary ».

- Euler's formula (1748).

- Complex plane: Caspar Wessel (1797), Jean Robert Argand (1806), John Warren (1828), Carl Friedrich Gauss (1831).

Towards Complex Numbers

The solution of the $\mathbb{R}$-equation

$$x^2 + 1 = 0$$

is not in $\mathbb{R}$

(since $x^2 > 0$ for any $x \in \mathbb{R}$).

We shall declare that the symbol

$$i$$

— called the ***imaginary unit*** —

is a solution of the equation

$$x^2 + 1 = 0,$$

which means that i is such that

$$i^2 = -1$$

or

$$i = \sqrt{-1}$$

Given the property $\sqrt{ab} = \sqrt{a}\sqrt{b}$,

the introduction of $i = \sqrt{-1}$ allows us to find

the square roots of any negative quantity.

For instance, $\sqrt{-4} = \sqrt{(-1)4} = \sqrt{-1}\sqrt{4} = i2$

At first, *i* is a « blind » or symbolic extension

deprived of any intuition.

How can we endow

the symbol *i*

with some form of intuitive understanding?

Besides using numbers to *count*, *order* and *measure*,

we can also use numbers to specify

how much of each « ingredient » is present

in a particular composition

(like the different ingredients in a cooking recipe),

In particular, we can use numbers

as ***scalars*** in a ***vector space***.

A digression on Vector Spaces

A ***k-vector space*** $(V, +, k = \mathbb{R} \text{ or } \mathbb{C})$

is an additive commutative group $(V, +)$

— a set V of vectors equipped with an invertible addition $+$

(and the corresponding neutral element, the null vector $\overrightarrow{0}$) —

equipped with a scalar multiplication

$$k \times V \to V$$

by elements of a field k ($\mathbb{R}$ or $\mathbb{C}$)

called *scalars*,

which refers to the fact that vectors can be « scaled » (e.g. stretched).

A k-vector space can be visualized

as a set V of « arrows »

based at a common origin

where vectors can be « scaled » by numbers in k and added,

such that the addition $\overrightarrow{v} + \overrightarrow{w}$ of two vectors $\overrightarrow{v}$ and $\overrightarrow{w}$

is the vector corresponding

to the diagonal of the parallelogram spanned by $\overrightarrow{v}$ and $\overrightarrow{w}$.

Linear combinations

Given two vectors $\overrightarrow{v}$ and $\overrightarrow{w}$,

we can define the ***linear combination***

$$a\overrightarrow{v} + b\overrightarrow{w},$$

where $a, b \in k$.

The vectors $\overrightarrow{v}$ and $\overrightarrow{w}$ are scaled

by the numbers a and b respectively

and the resulting scaled vectors $a\overrightarrow{v}$ and $b\overrightarrow{w}$ are added,

which yields the vector $a\overrightarrow{v} + b\overrightarrow{w}$.

Basis of a Vector Space

Given a vector space $(V, +, k)$,

it is possible to define a set of (non-collinear) vectors $(\vec{e}_1, \ldots, \vec{e}_n)$

— called a ***basis*** —

such that any vector v can be written

as a linear combination of the elements of the basis

$$\vec{v} = v_1 \vec{e}_1 + v_2 \vec{e}_2 + \ldots + v_n \vec{e}_n$$

The set of numbers $(v_1, v_2, \ldots, v_n)$ are the ***coordinates*** of $\vec{v}$ in the basis $(\vec{e}_1, \ldots, \vec{e}_n)$.

The number n of elements in the basis is the ***dimension*** of the vector space.

Real Vector Spaces

In a $\mathbb{R}$-vector space, multiplying a vector $\overrightarrow{v}$ with a scalar k

stretches ($|k| > 1$) or shrinks ($|k| < 1$) the vector

while changing ($k < 0$) or not ($k > 0$) its orientation.

In particular, multiplying by the scalar -1

keeps the length of the vector $\vec{v}$,

but changes its orientation.

In other terms, multiplying by -1

rotates the vector in $180°$ or π radians.

Scaling by $i = \sqrt{-1}$ or How to Factorize the Inversion of a Vector

Since $-1 = i \times i$, we have

$$(-1)\overrightarrow{v} = (ii)\overrightarrow{v} = i(i\overrightarrow{v})$$

(where we have used that in a vector space $(kk')v = k(k'v)$).

Since multiplying a vector $\overrightarrow{v}$ by -1 inverts the orientation of $\overrightarrow{v}$,

multiplying by i can be understood as a rotation in $90°$ or $\frac{\pi}{2}$ rad.

The imaginary unit i is a new number,

such that multiplying a vector by i

rotates the vector in 90°

(without stretching or shrinking its length).

Complexification as an Enhancement of Continuity

It is not possible to rescale a unit vector $\overrightarrow{1}$ on $\mathbb{R}$

by means of a continuously varying scalar $k(t)$

so that the transformed vector is $-\overrightarrow{1}$.

On the contrary,

$\overrightarrow{1}$ can be transformed into $-\overrightarrow{1}$

by means of a continuous rotation

obtained by multiplying $\overrightarrow{1}$

with a continuously varying complex scalar.

« [...] that curiously elegant form of continuity that is provided by *complex numbers* [...] Perhaps, in complex numbers will ultimately be found the resolution between the discrete and continuous. »

R. Penrose, Foreword to Schrödinger, *Nature and the Greeks*, p.x.

The field $\mathbb{C}$ of complex numbers

is an extension of $\mathbb{R}$ such that each « scalar » $z \in \mathbb{C}$

not only specifies a ***dilatation*** or ***stretching*** of vectors

by an amount r

but also a ***rotation*** in an angle θ.

Whereas all the previous number systems

encode the concept of ***magnitude***

(how much?, how many?, how big?),

the complex numbers enrich the concept of

magnitude with an ***orientation***.

Complex numbers can therefore be understood as

oriented magnitudes.

Complex numbers become pertinent

in situations in which we combine certain « ingredients »

in such a way that it is not only important

to specify *how much* of each ingredient is present,

but also the *orientation* of each ingredient

over a full circle $0 \leq \theta < 2\pi$.

Example: Fourier Synthesis

In Fourier synthesis,

circular motions are combined

to yield a periodic signal,

where the corresponding linear combination

requires to specify the amplitude of each motion

as well as their initial orientation.

Hence, the « scalars » will be complex numbers $z = (r, \theta)$,

where r will fix the amplitude of the corresponding component

and θ the initial orientation.

We can describe a complex number as a pair

$$z = (r, \theta),$$

where $0 \leq r$

is the *radius* (or *length* or *modulus*) of z

and $0 \leq \theta < 2\pi$

is the *angle* (or *argument*).

z « scales » vectors

by multiplying the length of $\overrightarrow{v}$ by r

and by rotating $\overrightarrow{v}$ counterclockwise in θ.

Since complex numbers depend

on two independent real variables

encoding a ***scaling*** of the length of vectors and a ***rotation***,

they are not geometrically schematized as points in a line (like $\mathbb{R}$),

but rather points in a plane,

the so-called ***complex plane***.

In order to identify each complex number z

with a point in a plane,

we can understand the plane

as a $\mathbb{C}$-vector space of dimension 1

in which the horizontal axis is identified with the real line $\mathbb{R}$.

This means that $\mathbb{C}$ is an extension of $\mathbb{R}$

as a plane is an extension of a line.

The real number 1 can then be identified with the vector $\overrightarrow{1}$

pointing from the origin to the number 1 on the real axis.

A complex number $z = (r, \theta)$ can then be identified

with the vector in the plane

obtained by « scaling » the vector $\overrightarrow{1}$ with $z = (r, \theta)$:

$$z \leftrightsquigarrow z\overrightarrow{1}.$$

Now, « scaling » the vector $\overrightarrow{1}$ by $z = (r, \theta)$

amounts to stretch $\overrightarrow{1}$ by a factor r and to rotate it in θ anti-clockwise.

Hence, the complex number $z = (r, \theta)$ can be identified

with a vector of length r rotated in θ from the real axis anti-clockwise.

In particular, the imaginary unit i

can be associated with the point in the plane

obtained by scaling the vector $\overrightarrow{1}$ with

the complex number z given by $r = 1$ and $\theta = \frac{\pi}{2}$,

which yields a vector pointing from the origin

to the number 1 on the vertical axis.

« If one formerly contemplated this subject from a false point of view and therefore found a mysterious darkness, this is in large part attributable to clumsy terminology. Had one not called $+1$, -1, $\sqrt{-1}$ positive, negative, or imaginary (or even impossible) units, but instead, say, direct, inverse, or lateral units, then there could scarcely have been talk of such darkness. »

C. F. Gauss, Theoria residuorum biquadraticorum. Commentatio secunda, *Commentationes Societatis Regiae Scientiarum Gottingensis Recentiores* 7 (1831), 89148.

The Complex Plane is in fact a « Line »

Since any vector in the complex plane

can be obtained by properly scaling $\overrightarrow{1}$

with a complex number z ,

the complex plane is a $\mathbb{C}$-vector space of dimension 1,

that is, as a line.

Complex Numbers in Cartesian Coordinates

We can now consider the complex plane as a $\mathbb{R}$-vector space of dimension 2.

By choosing the vectors $1 \leftrightsquigarrow (1,0)$ and $i \leftrightsquigarrow (0,1)$

as a basis we can express each z as a linear combination

$$z = a(1,0) + b(0,1)$$

where $a, b \in \mathbb{R}$ are the real coordinates

or (in short)

$$z = a + bi.$$

From Polar to Cartesian Coordinates

Since $a = rcos(\theta)$ and $b = rsin(\theta)$

the complex number $z = a + bi$ can be reexpressed

in terms of the polar coordinates (r, θ) as

$$z = r(cos(\theta) + isin(\theta))$$

or

$$z = re^{i\theta}$$

where we have used Euler's formula

$$e^{i\theta} = cos(\theta) + isin(\theta).$$

From Cartesian to Polar Coordinate

The polar coordinates of a complex number

$$z = a + bi$$

are given by the expressions

$$r = \sqrt{a^2 + b^2}$$

and

$$\theta = arctang(\frac{b}{a}).$$

Coordinates in the Complex Plane

A complex number z can be schematized

as a point in a Cartesian plane determined

either by ***Cartesian coordinates*** (a, b)

— in which case, $z = a + bi$ —

or in ***polar coordinates*** (r, θ)

— in which case, $z = re^{\theta}$ —

Complex Conjugate

The complex conjugate of a complex number

$$z = a + bi$$

is $\bar{z} = a - bi$,

which is the specular reflection of z

with respect to the real axis.

Algebraic properties of $\mathbb{C}$

- Complex number can be added:

$$(a + bi) + (c + di) = (a + b) + (b + d)i$$

where the additive inverse of $a + bi$ is $-a - bi$.

- Complex number can be multiplied:

$$(a + bi) \times (c + di) = ac + adi + bic + bdi^2 = (ac - bd) + (ad + bc)i$$

where the multiplicative inverse of z is

$$\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2}.$$

$\mathbb{C}$ is a *field*.

Euler's Formula

$$e^{i\theta} = cos(\theta) + isin(\theta).$$

$$e^{i\pi} + 1 = 0$$

When exponential growth

becomes purely imaginary,

be sure you've entered a loop.

Properties of $\mathbb{C}$
I. The Fundamental Theorem of Algebra

Every polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.

Properties of $\mathbb{C}$
II. $\mathbb{C}$ is not endowed with a linear order.

Complex numbers do not entirely fall

within the scope of the definition of magnitude as

"that which can be *increased* or *decreased*"

(G.W.F. Hegel, *Encyclopedia of the Philosophical Sciences in Basic Outline*, §99).

Indeed, a complex number can vary

only with respect to its orientation, i.e. by keeping its radius constant.

Are complex numbers really numbers?

« Mais les nombres complexes […] sont-ils des nombres ? On peut, je crois, raisonnablement soutenir qu'à partir du moment où nous quittons toute 'linéarité', quand nous abandonnons la dimension 1, nous avons affaire à des constructions *à partir* des Nombres plutôt qu'à des Nombres. Au fond, l'essence intime des complexes est géométrique, c'est le 'plan complexe' qui délivre la vérité de ces 'nombres'. […] Ce sont en fait des nombres *qui ne nombrent pas* […] Il me paraît en outre déraisonnable de parler de 'nombres' quand il n'est pas même possible, en liaison avec le champ opératoire considéré de dire que tel 'nombre' est plus grand ou plus petit qu'un autre. · Bref : un corps *de nombres* doit à mon avis être un corps *ordonné*, ce que ne sont [pas] les complexes […] Finalement, je restreindrai le concept de Nombre […] à ce qui se laisse déployer selon l'intuition d'une ligne.»

A. Badiou, *Le Nombre et les nombres*, *op.cit.*, p.278.

Summary of Number Systems

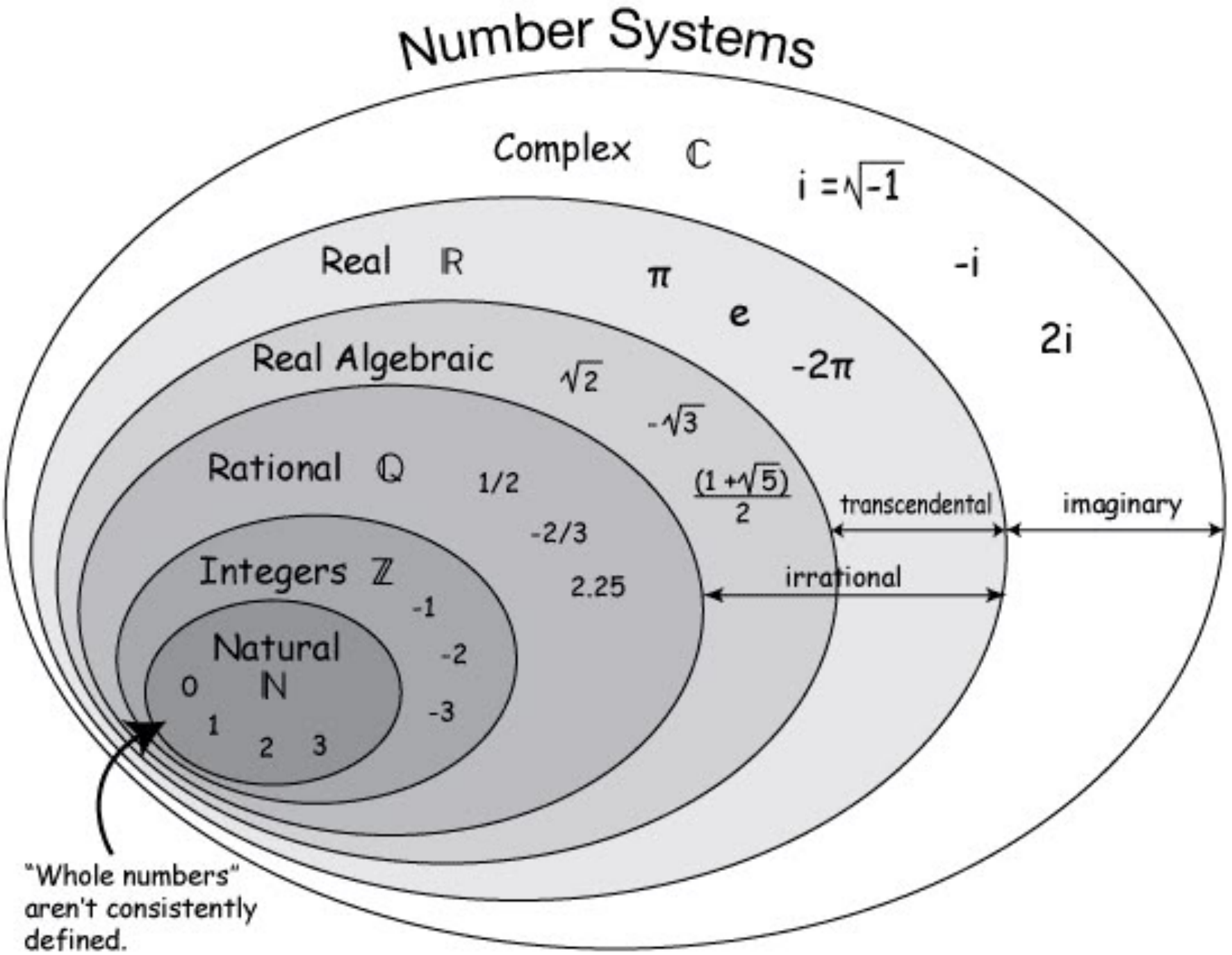
- Natural numbers $\mathbb{N}$
- Entire numbers $\mathbb{Z}$
- Rational numbers $\mathbb{Q}$
- Irrational numbers:

    .algebraic

    .transcendental :

        .computable,
        .non-computable.

- Real numbers $\mathbb{R}$.
- Complex numbers $\mathbb{C}$.


Number Systems
Complex ℂ
i = √-1
-i
2i
Real ℝ
π
e
-2π
Real Algebraic
√2
-√3
Rational ℚ
1/2
(1+√5)/2
-2/3
2.25
transcendental
imaginary
irrational
Integers ℤ
-1
-2
-3
Natural
ℕ
0
1
2
3
"Whole numbers" aren't consistently defined.

Generalizations in the Evolution of Number Systems

- From the ***finite*** to the ***infinite***.
- From the ***discrete*** to the ***continuum***.
- From ***magnitudes*** to ***oriented magnitudes***.

From the Relative to the Absolute

The Algebra-Geometry Duality

In the XVIIth century, Fermat and Descartes

independently invented analytic geometry,

that is, the study of the correspondence

between *algebraic expressions* and *geometric figures*

by using ambient spaces endowed with coordinate systems.

Ref.: C.B. Boyer, *History of Analytic Geometry*, (2004) [1956], Dover Publications,

From the algebraic equation at stake,

one can define a equation

and the solutions of the equation

can be « schematized »

as points in a coordinate space.

Examples

- The algebraic expression $x - 1$ defines the equation $x - 1 = 0$ whose unique solution is $x = 1$, which can be schematized as a point on a line in which a coordinate system has been defined.

- The algebraic expression $y - 2x$ defines the equation $y - 2x = 0$ whose unique solution is a set of pairs (x, y), each of which defines a point in a plane encoded with coordinates. The resulting geometric figure is a straight line passing by the origin of the coordinate system.

Conics

Conics

— i.e. elipses, circles, parabolas and hyperbolas —

are geometric figures in the plane

associated to algebraic expressions of the form

$$ax^2 + bxy + cy^2 + dx + ey + f$$

What is the geometric figure that « schematizes »

the expression $x^2 + y^2 - 1$?

In other terms, what are the solutions to the equation

of a circle of radius 1

$x^2 + y^2 = 1$?

Transcendental Relativity

The geometric ‘figure’ that ‘schematizes’ a given algebraic expression,

far from being absolutely defined,

depends on the kind of numbers

in which one searches for the solutions of the corresponding equation.

The solution to a problem depends

on the « transcendental framework »

— in the Kantian sense of the term —

in which one searches for such a solution.

Solutions to the Circle Equation

- Solutions in $\mathbb{Z}^2$ (entire circle)

$$\{(0,1),(0,-1),(1,0),(-1,0)\};$$

- Solutions in $\mathbb{Q}^2$ (rational circle):

$$x = \frac{q^2-1}{q^2+1}; y = \frac{2q}{q^2+1}, \text{ with } q \in \mathbb{Q}$$

- Solutions in $\mathbb{R}^2$ (real circle)

$$x = cos(\theta); y = sin(\theta), \text{ with } \theta \in \mathbb{R}.$$

- Solutions in $\mathbb{C}^2$ (complex circle).

« [...] the unit circle is the set of all solutions in the x-y plane to the equation

$$x^2 + y^2 = 1.$$

The set of solutions of this equation depends on the type of number that one understands x and y to refer to. [...] There are many more abstract types of numbers [than the integers, the rationals, the real, and the complex numbers] and each gives rise to its own circle. In the classical set theoretic framework, there is thus an integer circle, a rational circle, a real circle, a complex circle and myriads of others […] »

M. Barr, C. McLarty, C., & C. Wells, *Variable Set Theory*, 1986.

Such a transcendental relativity

was sublated by A. Grothendieck's introduction

of the notion of ***functor of solutions***

in the framework of his refoundation

of *algebraic geometry*.

Ref. A. Grothendieck & J.A. Dieudonné, *Éléments de Géométrie Algébrique I*, 1971, Springer.

Each transcendental framework A

is an algebraic structure

— a ring or a k-algebra —

such that it makes sense to replace the x

in the equations with element in A.

A ***transformation*** between different transcendantales frames

(also called a *morphism*)

$$f : A \rightarrow B$$

is a function between the corresponding sets of numbers

that preserve the algebraic structure (e.g. the ring structure),

that is, a ***structure preserving function***.

Morphisms Between Rings

Given two rings A and B, a morphism between them is a function

$$f : A \to B$$

that preserves the ring structure, that is:

- Addition:

$$f(a + a') = f(a) + f(a').$$

- Multiplication:

$$f(a \times a') = f(a) \times f(a').$$

- Neutral element for the multiplication:

$$f(1_A) = 1_B.$$

Note: It follows from these conditions that the additive inverses and the neutral element for the addition are also preserved.

Transcendental Parallax

A morphism

$$f : A \to B$$

between different « transcendental frames »

— i.e. a structure-preserving function between the underlying sets —

sends solutions to solutions.

For instance, if $a, b \in A$ solve the circle equation

$$x^2 + y^2 = 1$$

and if

$$f : A \to B$$

is a morphism of rings,

then $f(a)$ and $f(b)$ are elements in B

that also solve the circle equation.

Indeed,

$$f(a)^2 + f(b)^2 = f(a)f(a) + f(b)f(b) = f(a^2) + f(b^2) = f(a^2 + b^2) = f(1) = 1$$

Grothendieck's key idea

is that the 'solution' of a polynomial equation

$$p(x) = 0$$

is not the set of its roots in some privileged transcendental frame,

but rather the functorial 'application' that assigns

a different set of solutions to each possible transcendental frame A

in such a way that it takes into account the « parallax effects »

associated to transformations between different transcendental frames or perspectives.

Typically, problems of transcendental relativity have been addressed either by:

- Searching for invariants (***invariantism***);
- Canonizing or hypostasizing a given transcendental perspective (***provincialism***).

Grothendieck's innovative solution

to the problem of transcendental relativity

amounts to consider all the perspectives at once

by taking into account the parallax deformations

arising from transformations

among the different transcendental standpoints.

« […] le signe d'une intelligence chamanique de première ligne est la capacité à voir simultanément selon deux perspectives incompatibles. »

E. Viveiros de Castro, *Métaphysiques cannibales,* 2009, Paris: PUF.

The solution to the circle equation

can be described as a ***variable set*** V_{S^1}

that assignes a set of solutions $V_{S^1}(A)$

to each transcendental frame A

that co-varies (or varies « covariantly »)

when the transcendental frame is changed.

The Functor of Solutions

In category-theoretical terms,

such a variable set is given by a (covariant) ***functor***

$$V_{S^1} : Ring \to Set$$

from the category of rings *Ring*

to the category of sets *Set* such that

- $V_{S^1}(A)$ is the set of solutions of the circle equation in the ring A;

- $V_{S^1}(f) : V_{S^1}(A) \to V_{S^1}(B)$, where $f : A \to B$ a morphism of rings, is a function between the solutions in A and the solutions in B induced by the morphism of rings f.

The functor of solutions

associated to an algebraic expression

encodes all the corresponding solutions

in all possible transcendental frames

as well as the « parallax effects »

associated to transformations between transcendental frames.

By considering non-standard solutions

in well-chosen transcendental frames,

Grothendieck managed in describing (for instance)

differential/infinitesimal structures

associated to geometric figures.

Infinitesimals as Nilpotent Elements

For $a < 1$ small,

a^2 is even smaller $a << 1$.

Hence Grothendieck associates infinitesimals

with nilpotent non-zero elements $x \in A$ such that

$$x^2 = 0.$$

Solutions Given by Dual Numbers

Let's consider the $\mathbb{C}$-algebra

$$A = \mathbb{C}[\epsilon]/(\epsilon^2),$$

whose elements are polynomials of the form

$$p(\epsilon) = a + b\epsilon$$

subjected to the condition $\epsilon^2 = 0$.

Now, the pair $(1,\epsilon) \in A^2$ is a solution to the circle equation:

$$1^2 + \epsilon^2 = 1.$$

Generic Solution with the Tangent Vector

Let's consider a solution of the form $(x_0 + u\epsilon, y_0 + v\epsilon) \in A^2$.

In order for this to be a solution we have

$$(x_0 + u\epsilon)^2 + (y_0 + v\epsilon)^2 = 1$$

By expanding, we find the equations

$$x_0^2 + y_0^2 = 1$$

— which defines a point of the circle —

$$x_0 u + y_0 v = 0$$

—which defines a tangent vector perpendicular to the position vector —

The solution $(x_0 + u\epsilon, y_0 + v\epsilon) \in A^2$ can be rewritten as

$$(x_0, y_0) + \epsilon(u, v)$$

i.e., as a vector (x_0, y_0) whose head is a point in the real circle

plus a vector $\epsilon(u, v)$ of infinitesimal length

—- since $\|\epsilon(u, v)\| = \epsilon\sqrt{u^2 + v^2}$) —-

perpendicular to (x_0, y_0).

The tangent vector is represented by the nilpotent part of the coordinate.

For instance, the solution $(1,\epsilon) \in A^2$ represents a point in the circle

whose coordinates are (1,0) enriched with a tangent vector.

Solutions in a Finite Field

The finite field $\mathbb{F}_3 = \{0,1,2\}$ is defined by

$a = b \bmod 3$ if $a = 3n + b$ where $n \in \mathbb{N}$,

which implies that

$$4 = 1$$

Hence (2,0) and (0,2) are solutions of the circle equation.

Indeed,

$$2^2 + 0^2 = 4 = 1$$

Thanks

for your attention !!!

Appendices

Inner Products in Vector Spaces

Vector Spaces Equipped with Inner Products

A vector space $(V, +, k)$ can be equipped

with an additional structure called ***inner product*** $\cdot$,

which is a k-valued multiplication of vectors:

$$\langle \cdot , \cdot \rangle : V \times V \to k$$

which is

- Conjugate symmetry $\langle \vec{v}, \vec{w} \rangle = \overline{\langle \vec{v}, \vec{w} \rangle}$ which implies $\langle \vec{v}, \vec{v} \rangle \in \mathbb{R}$.
- Positive-definite $\langle \vec{v}, \vec{v} \rangle > 0$ and $\langle \vec{v}, \vec{v} \rangle = 0$ if and only if $\vec{v} = 0$.
- The ***norm*** (length) of a vector is defined as $\| \vec{v} \| = \sqrt{\langle \vec{v}, \vec{v} \rangle}$.

$$\langle \vec{v}, \vec{w} \rangle = \|\vec{v}\| \|\vec{w}\| cos(\theta),$$

where θ is the angle between the two vectors.

Hence, the inner product encodes the product of the length $\|\vec{v}\|$

with the length of the protection of $\vec{w}$ in the direction of $\vec{v}$ (or viceversa).

$$\langle \vec{v}, \vec{w} \rangle = 0$$

when $\vec{v}$ and $\vec{w}$ are perpendicular or orthogonal

$$\text{(since } cos(\frac{\pi}{2}) = 0).$$

Euclidean Inner Product

The Euclidean inner product between two vectors

$$\overrightarrow{v} = v_1 \overrightarrow{e}_1 + v_2 \overrightarrow{e}_2 + \ldots + v_n \overrightarrow{e}_n$$

and

$$\overrightarrow{w} = w_1 \overrightarrow{e}_1 + w_2 \overrightarrow{e}_2 + \ldots + w_n \overrightarrow{e}_n$$

is given by

$$\langle \overrightarrow{v}, \overrightarrow{w} \rangle = v_1 \overline{w}_1 + v_2 \overline{w}_2 + \ldots + v_n \overline{w}_n$$

The norm of a vector is then given by the usual Pythagoras formula:

$$\| \overrightarrow{v} \| = \sqrt{\langle \overrightarrow{v}, \overrightarrow{v} \rangle} = \sqrt{v_1 \overline{v}_1 + v_2 \overline{v}_2 + \ldots + v_n \overline{v}_n} = \sqrt{|v_1|^2 + |v_2|^2 + \ldots + |v_n|^2}$$

The Number e

The Number e
I. Simple Interest

By depositing a capital C

with an annual rate of interest r written as a decimal number

(e.g. $r = 0.2$ for a 20 % of interest)

the bank gives back

an increased capital C' given by

$$C' = C + Cr = C(1 + r)$$

after a year.

The Number e
II. Compound Interest

If the annual rate is r and the year is divided in n parts,

the bank will give back

$$C' = C + C\frac{r}{n} = C(1 + \frac{r}{n}) \text{ after the first } n\text{-part of the year}$$

$$C'' = C'(1 + \frac{r}{n}) = C(1 + \frac{r}{n})^2 \text{ after the second } n\text{-part of the year}$$

$$\text{and } \widetilde{C} = C(1 + \frac{r}{n})^n \text{ after the last } n \text{ part of the year.}$$

The Number e
III. Compound Interest

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = 2.71828...$$

is the factor by which the initial capital is multiplied

after a year with an annual interest of 100%.

Instead of getting twice the initial capital,

one gets 2.71828... the initial capital.

k-algebras

Given a polynomial equation

$$p(x) = 0$$

over a *commutative ring with unity* k,

we can search for its solutions

in any ***k-algebra.***

k-algebras

A k-algebra

is a ring A

endowed with an external multiplication

$$k \times A \to A$$

by elements of the field k

(or a k-vector space endowed with an internal multiplication).

k-algebras

A ***k-algebra*** A is the minimal algebraic structure

such that the substitution of its elements $a \in k$

within a polynomial expression $p(x)$ makes sense,

which requires A to be equipped

with an addition, a multiplication

and a scalar multiplication with elements in k.